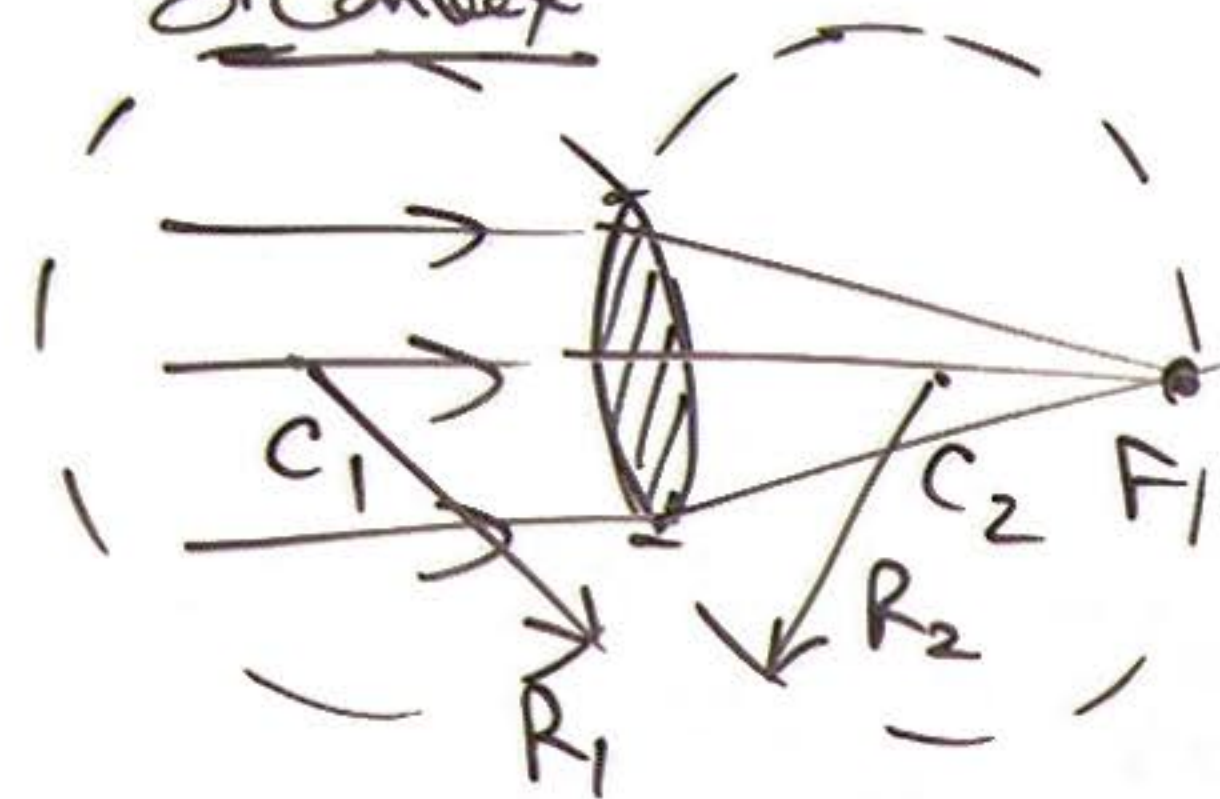


① Thin Lenses:

Focal length of a thin lens
is independent from dir of incidence.

Convex Lenses
(Converging)

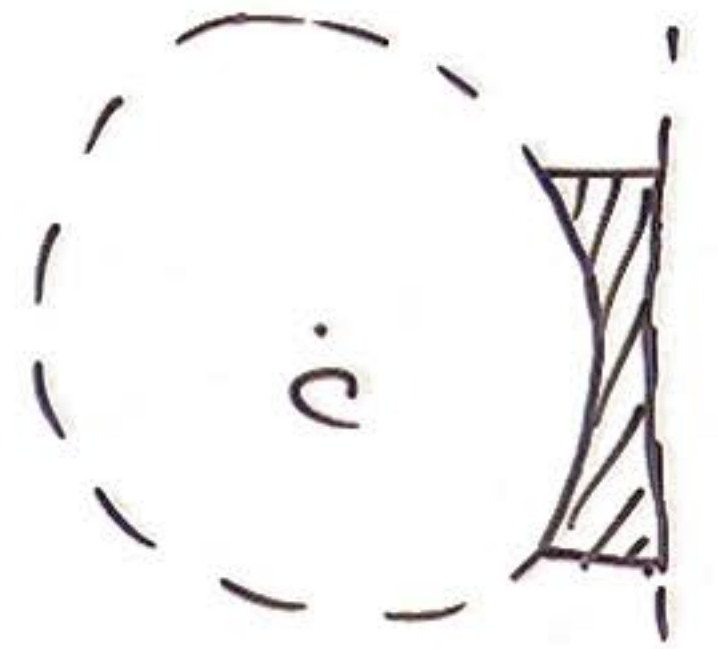
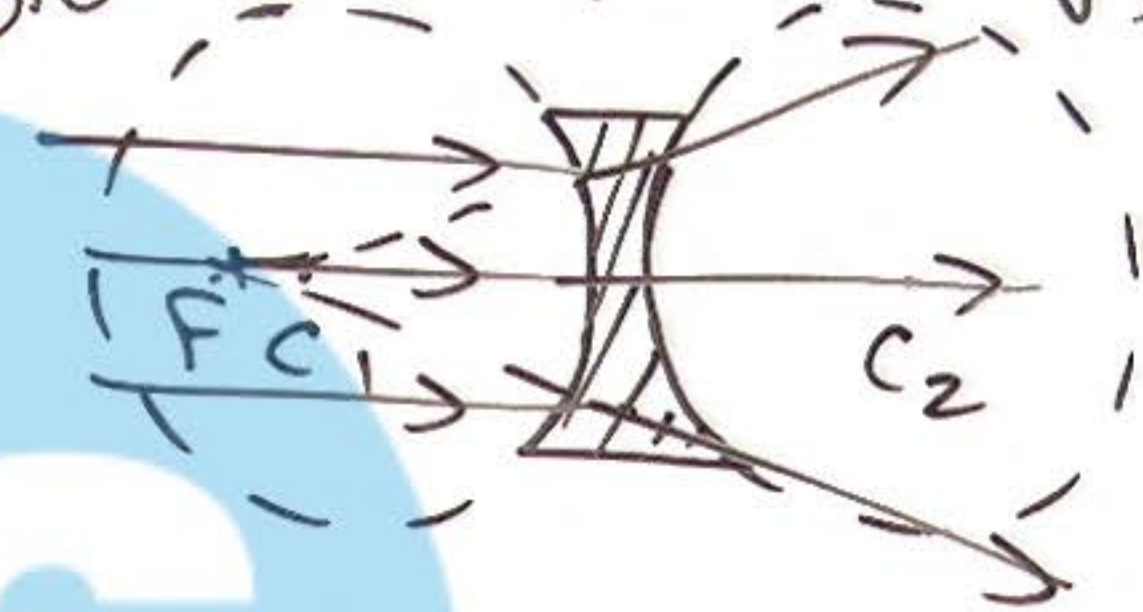


Concavo Convex.

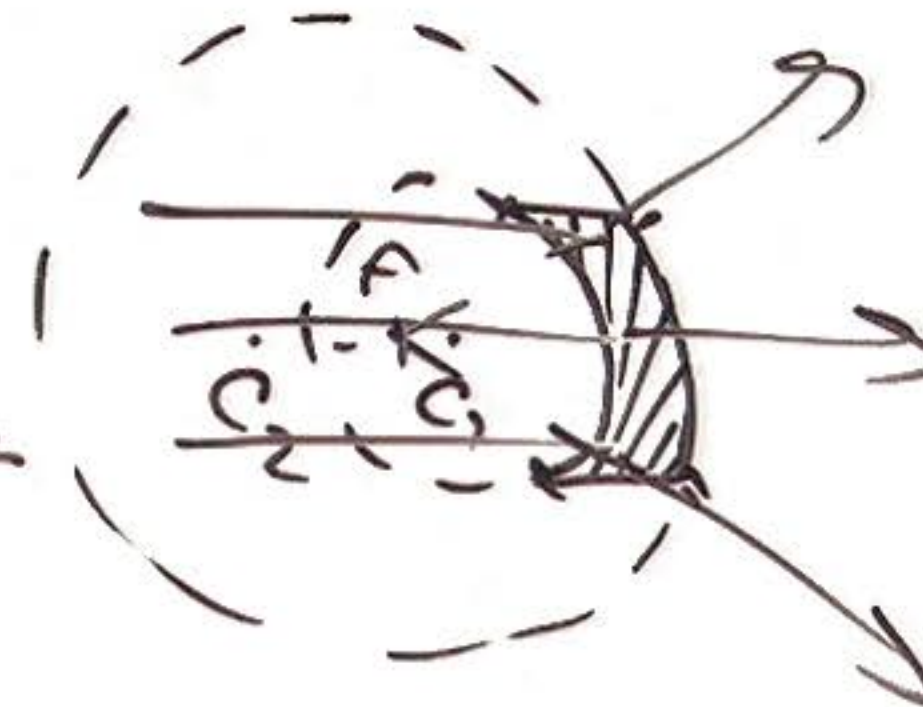


Plano Convex

Concave Lenses
(Diverging)



Plano Concave

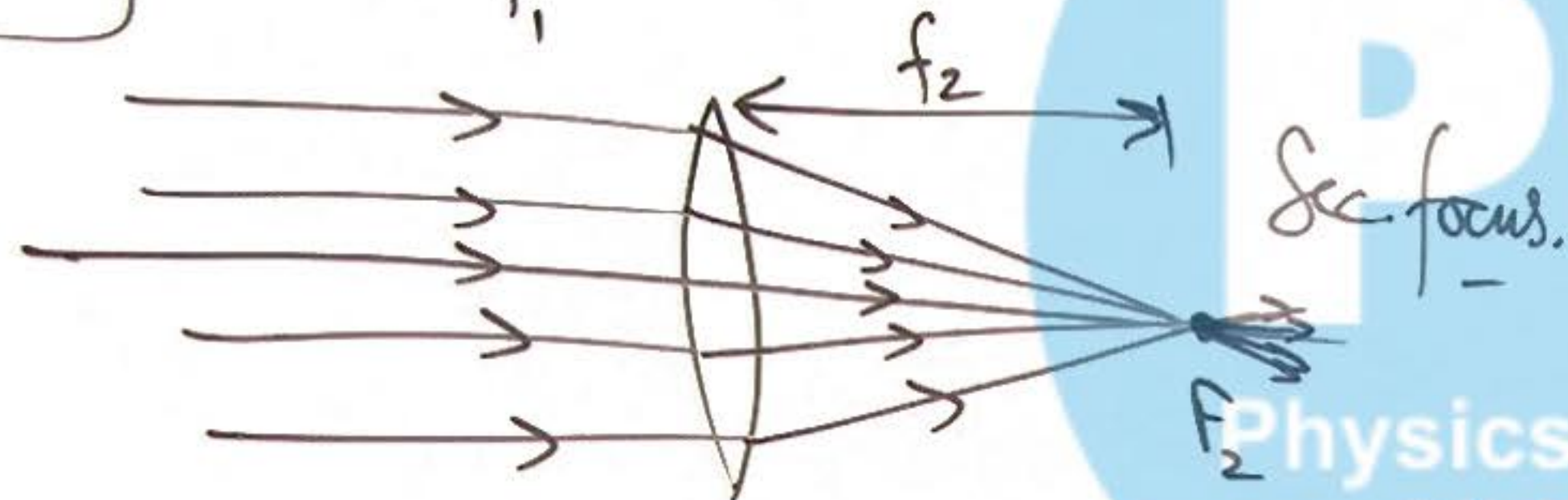
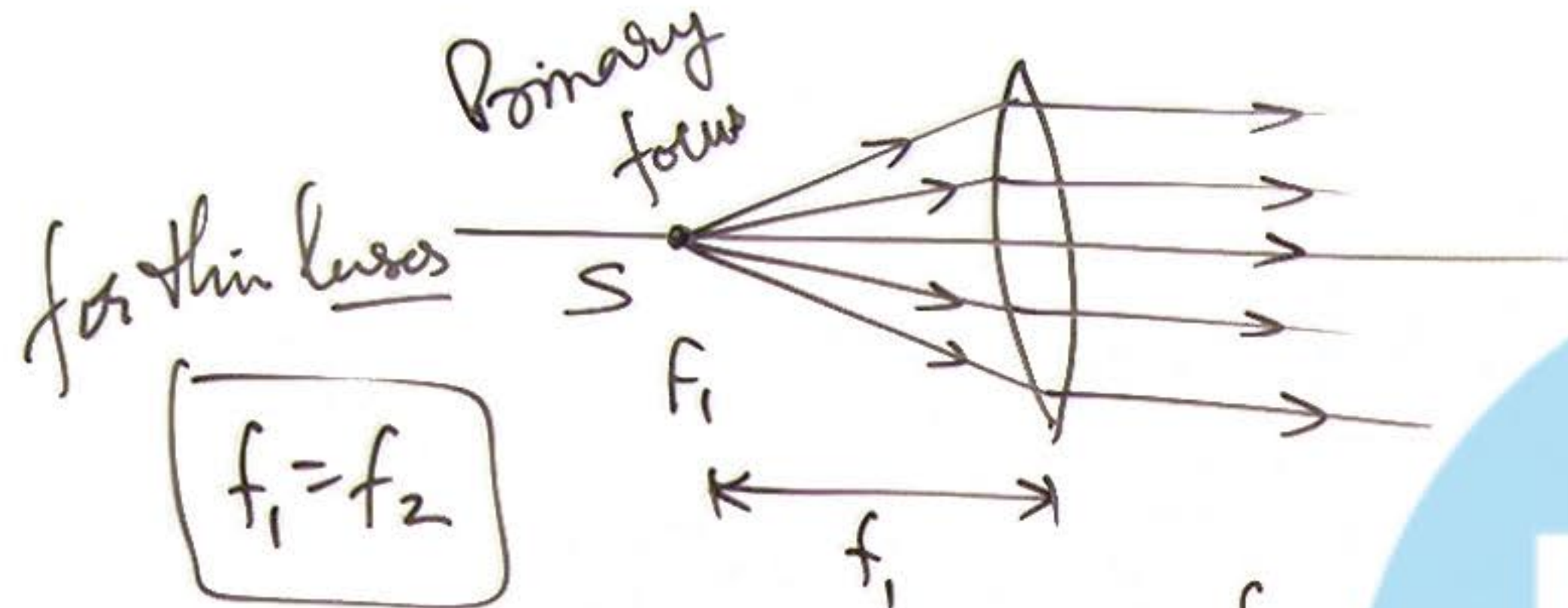


Convexo Concave.

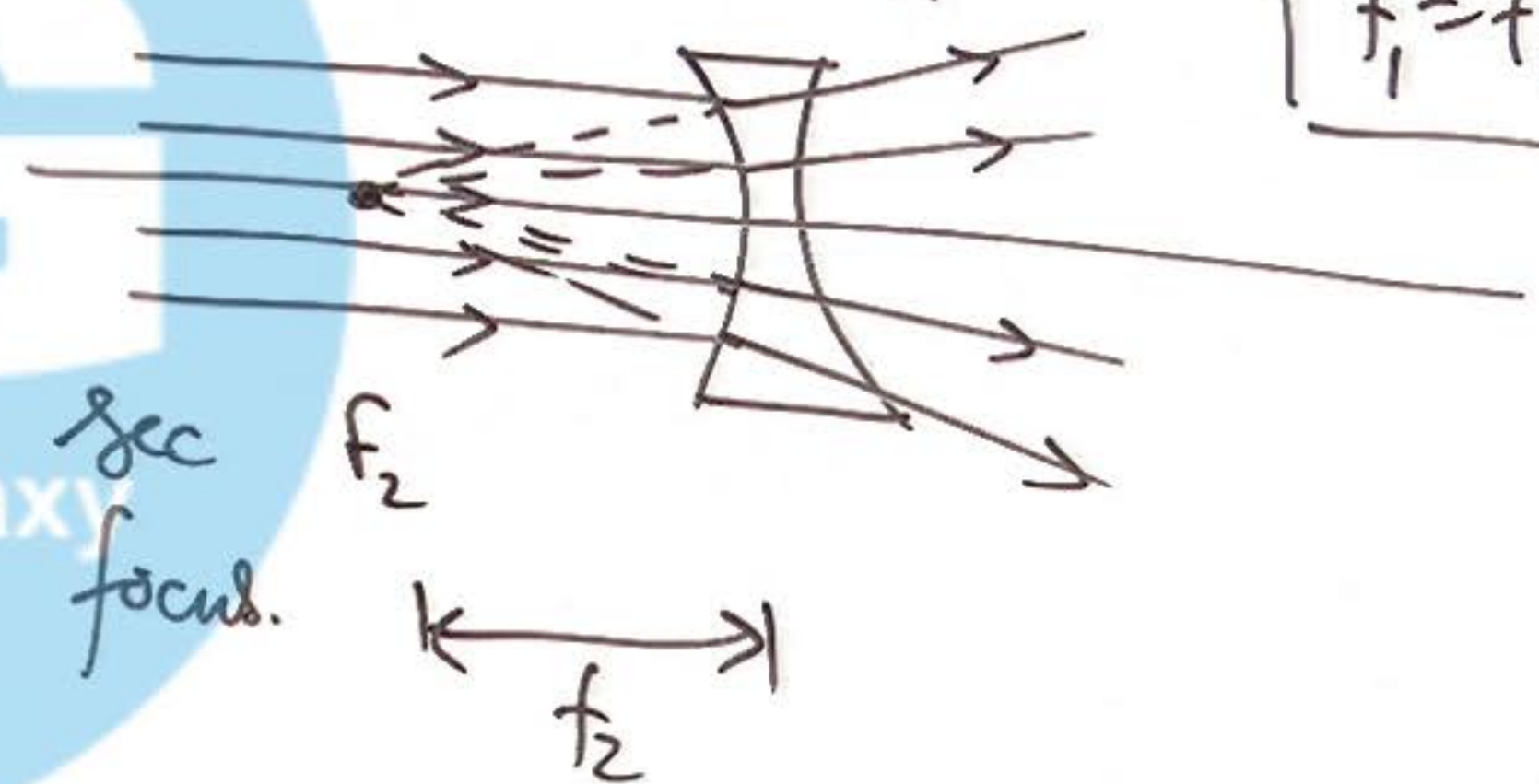
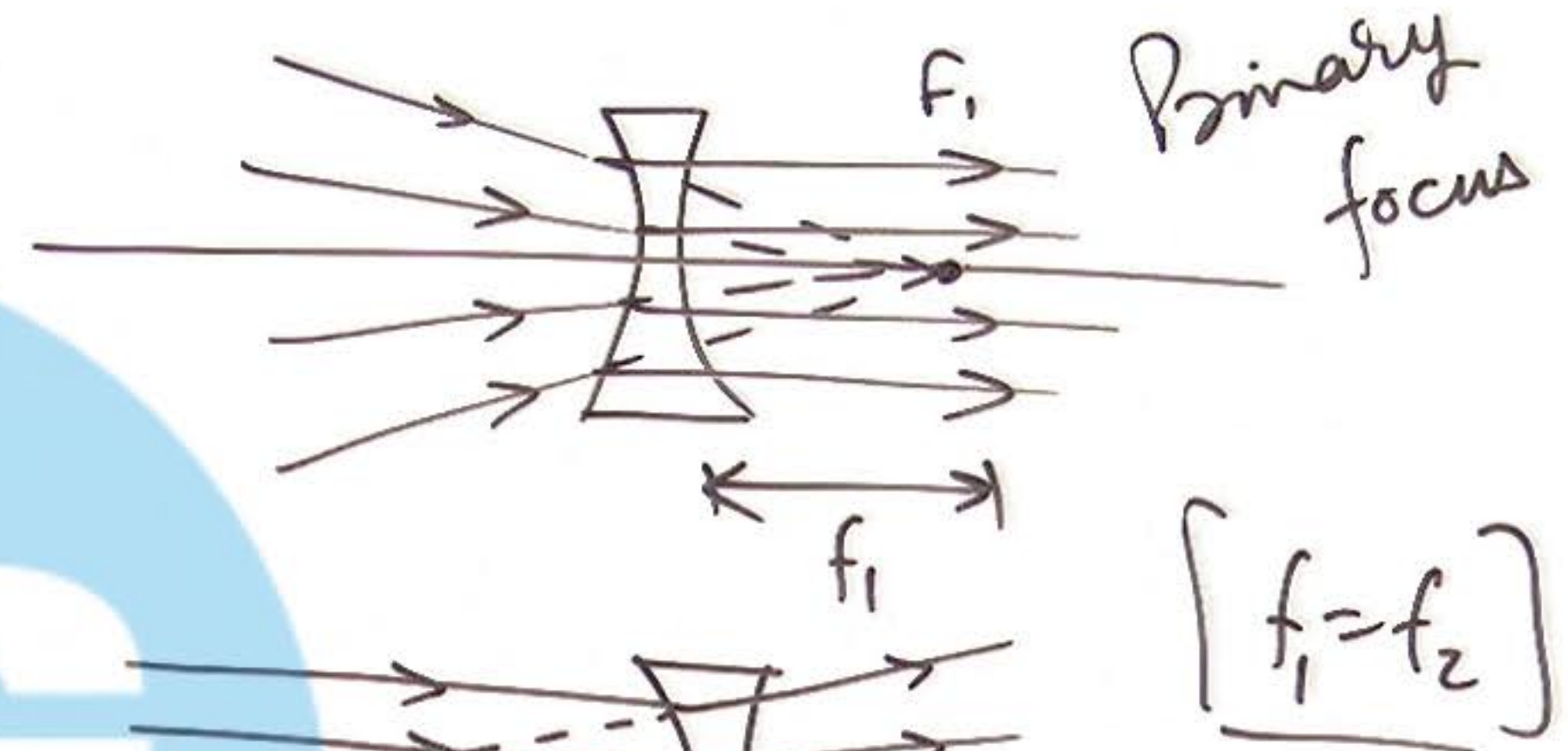
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② Primary & Secondary focus of a lens:

Convex



Concave

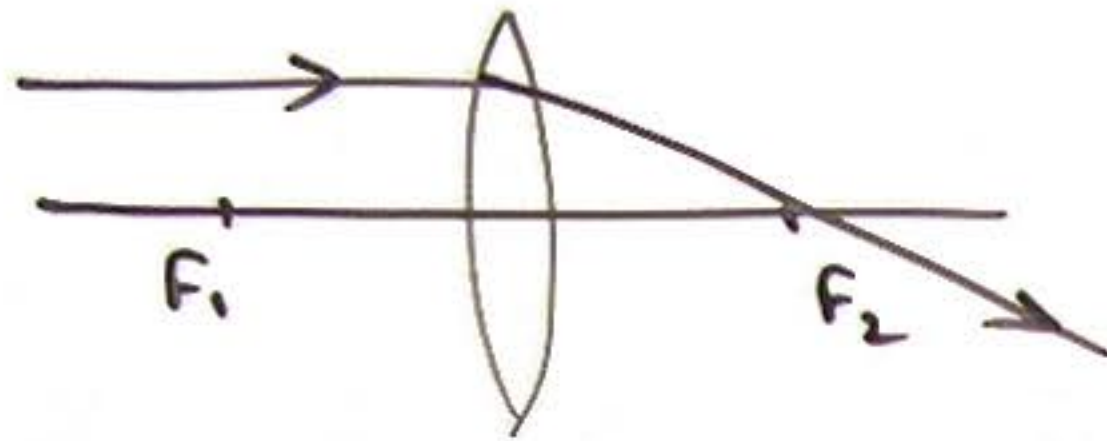


PG

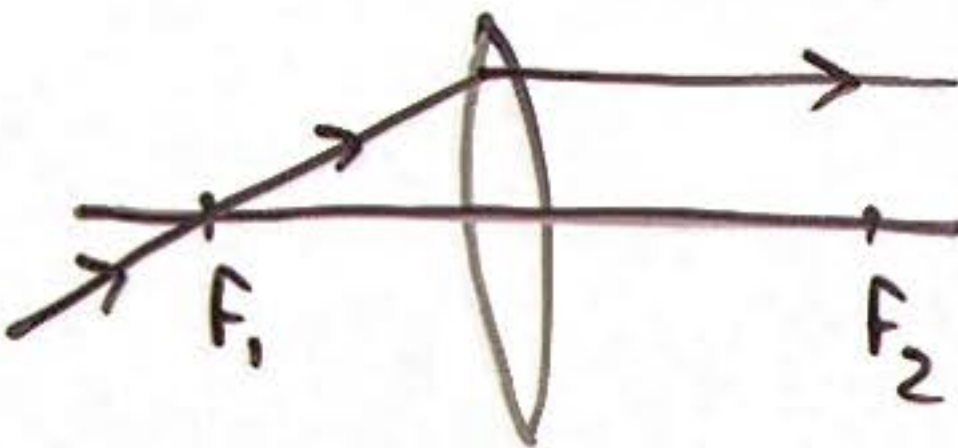
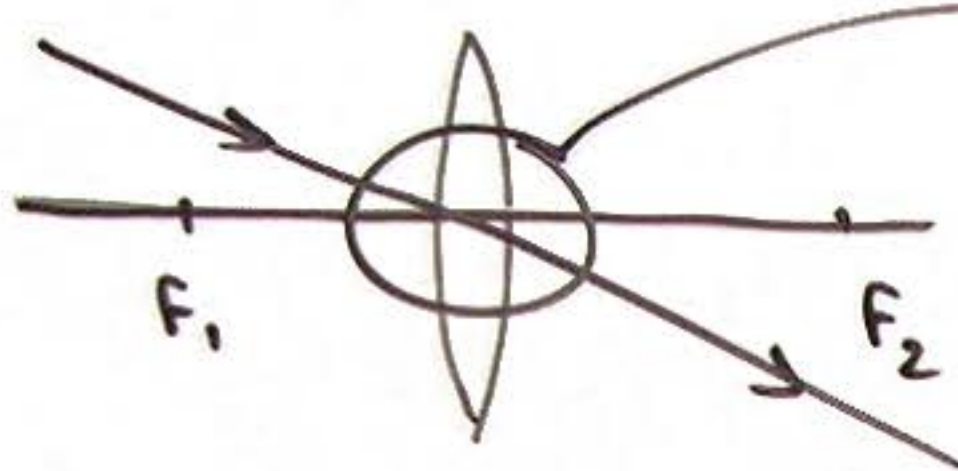
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② Refraction of Std rays Through Thin Lenses:

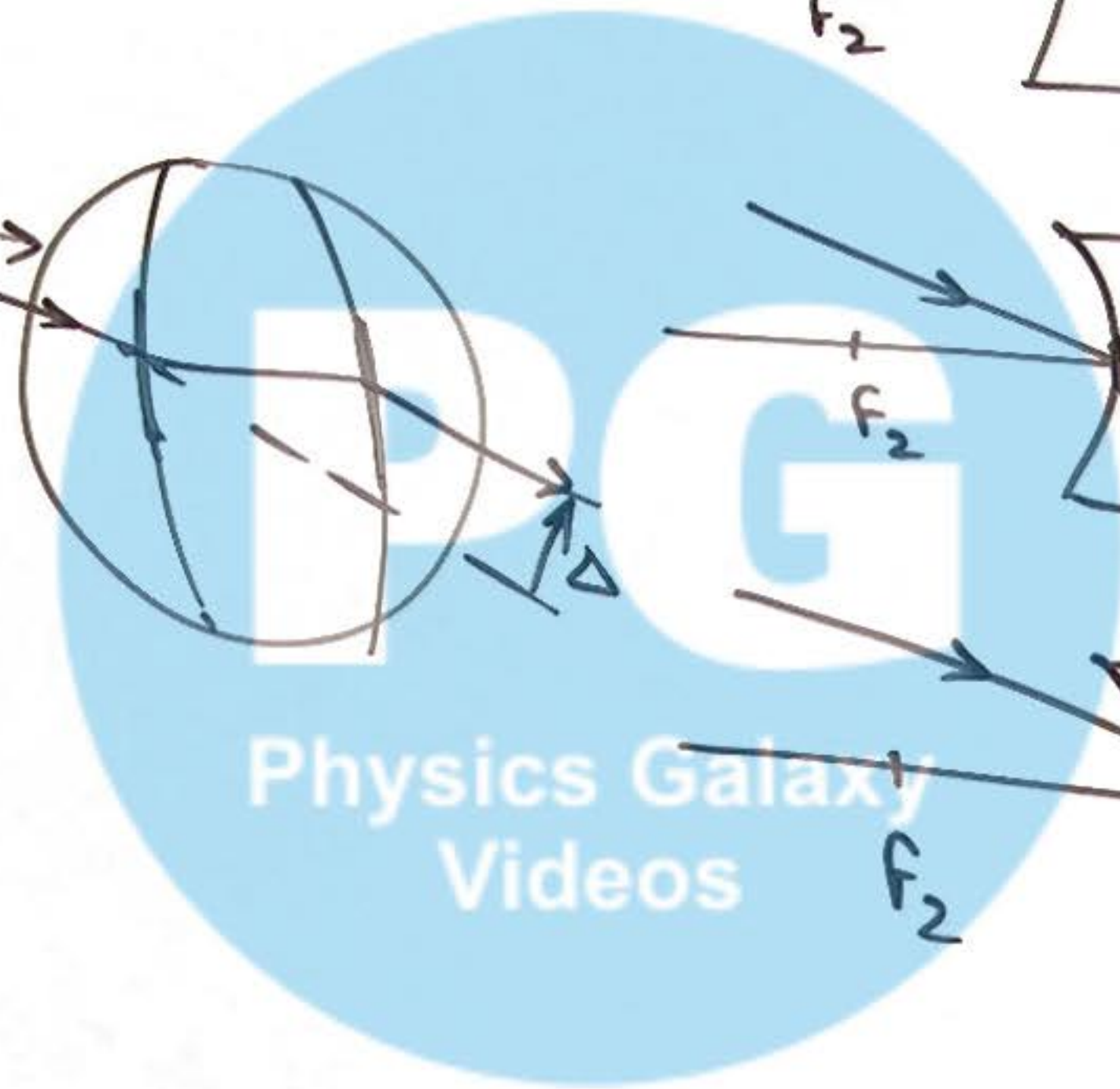
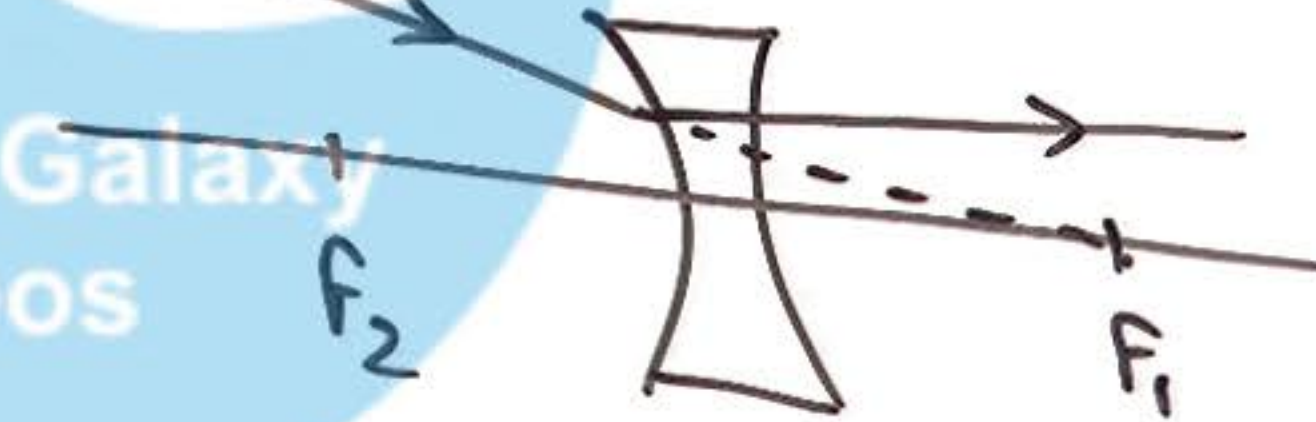
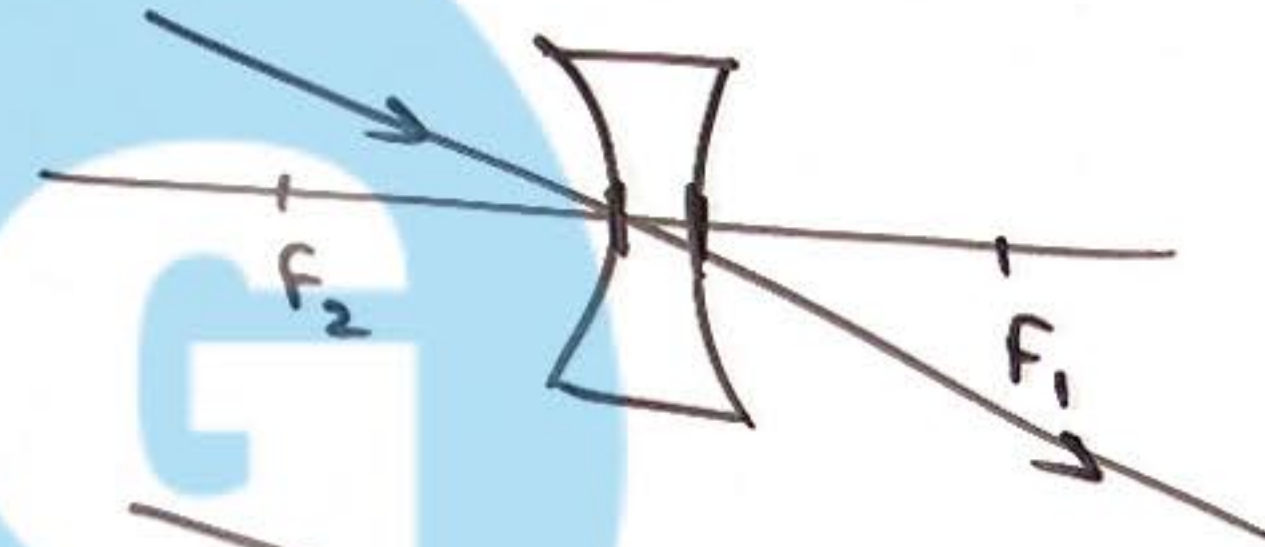
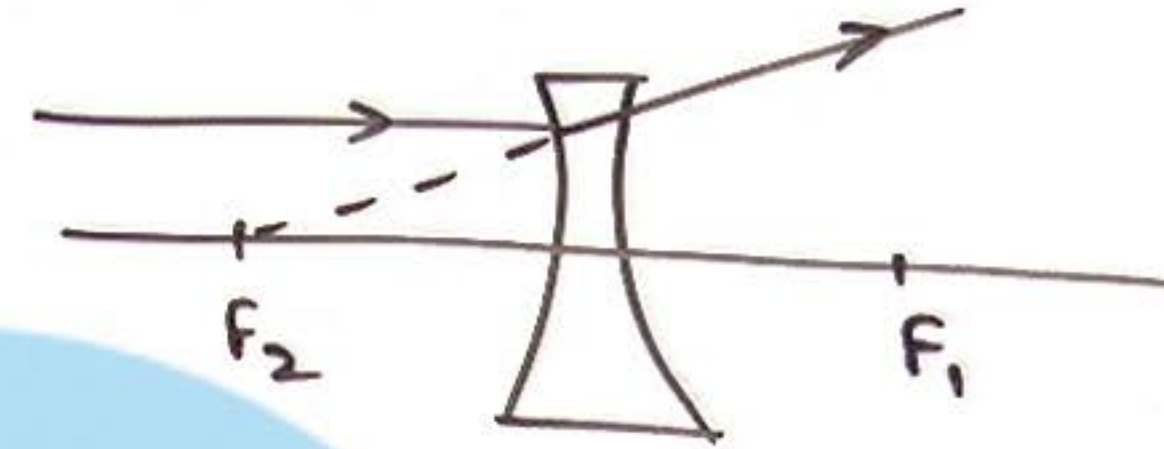
Convex



✓



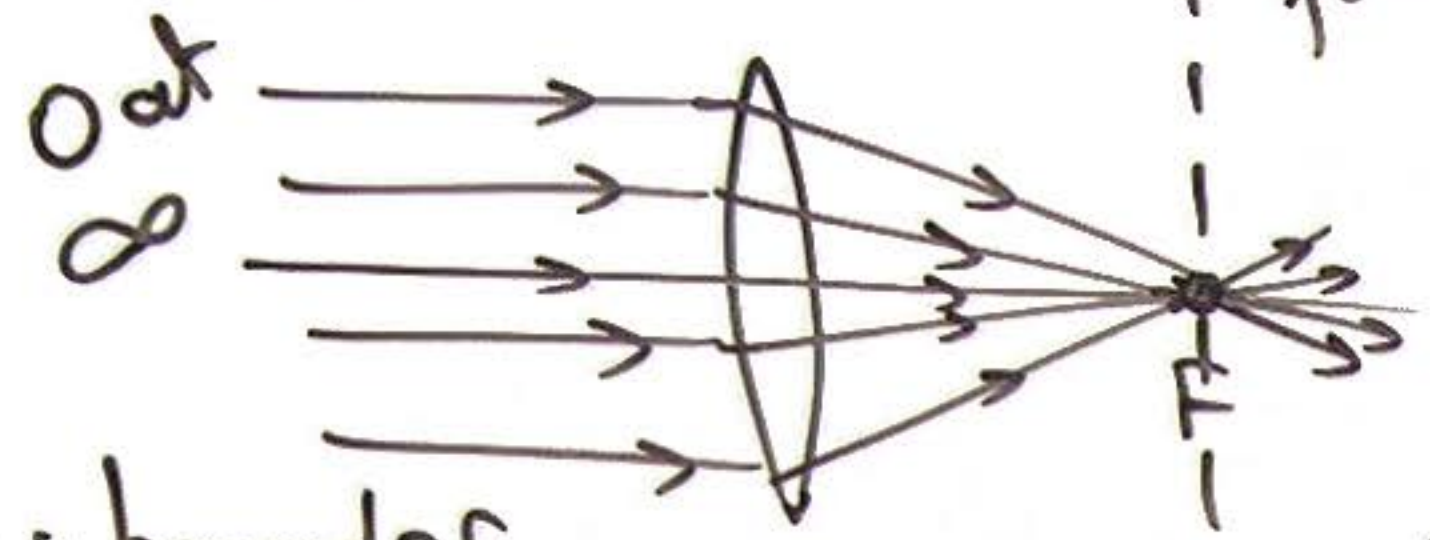
Concave



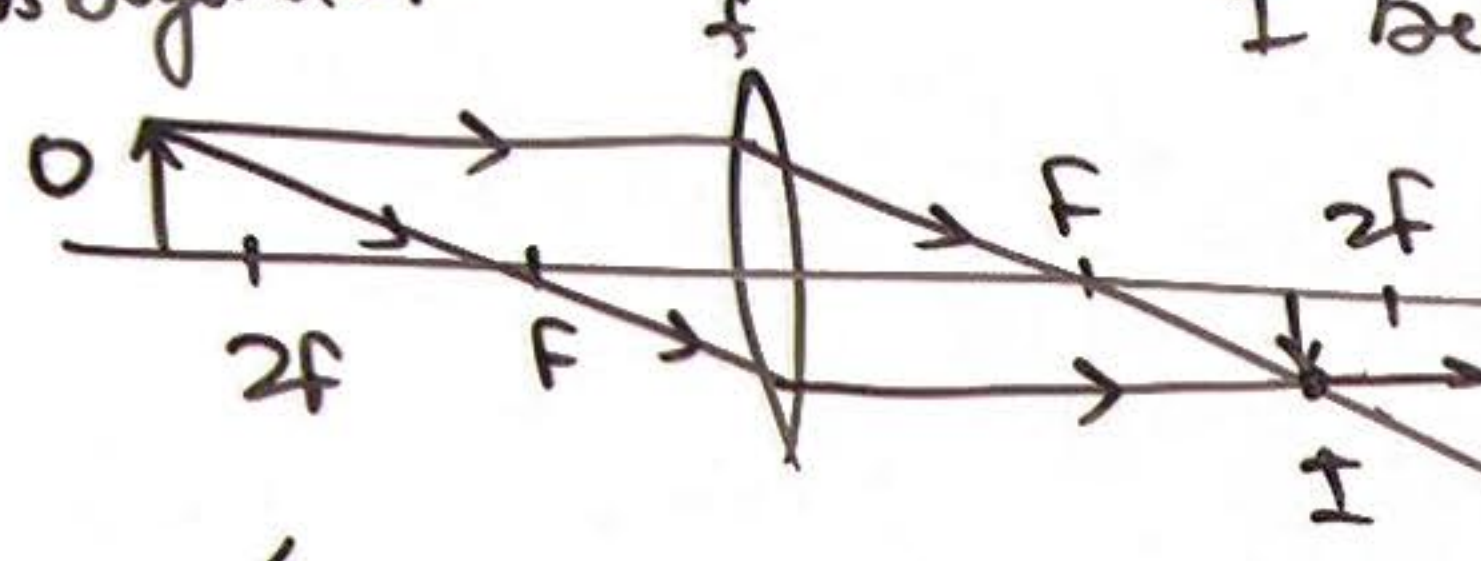
④ Relative position of Images produced by thin Lenses:

Case-I:
Convex lenses

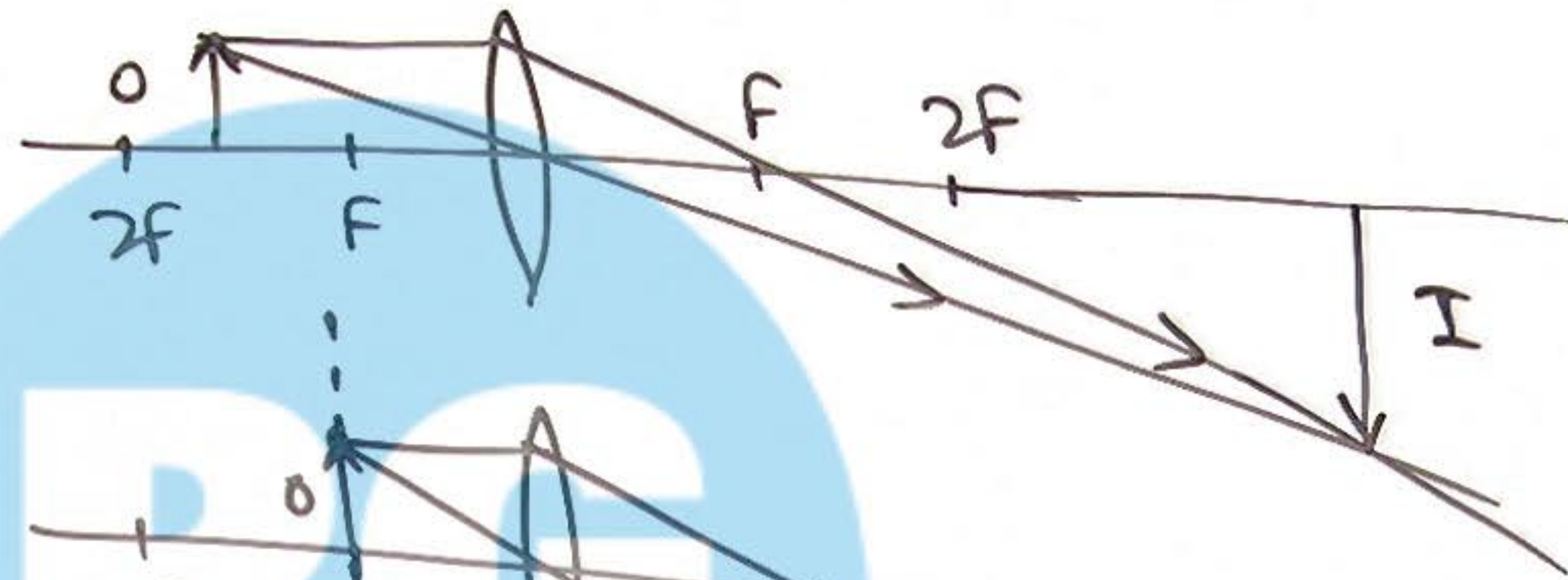
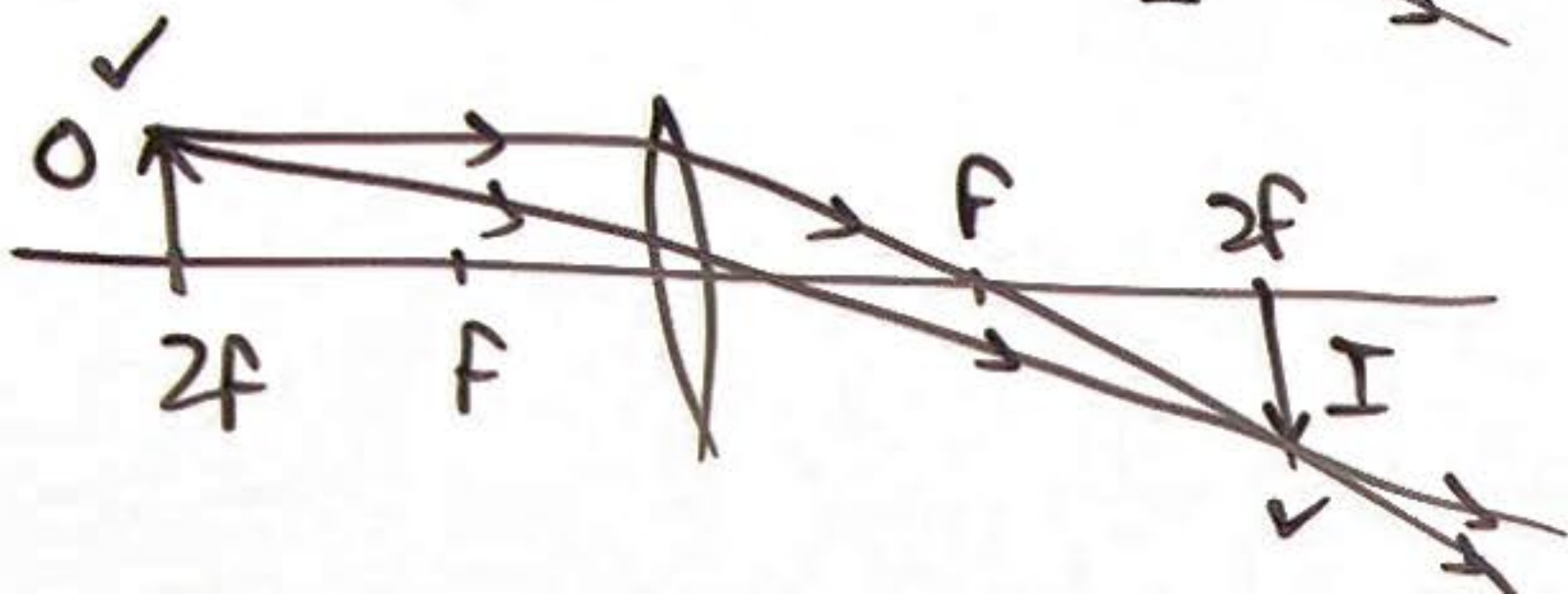
focal plane.



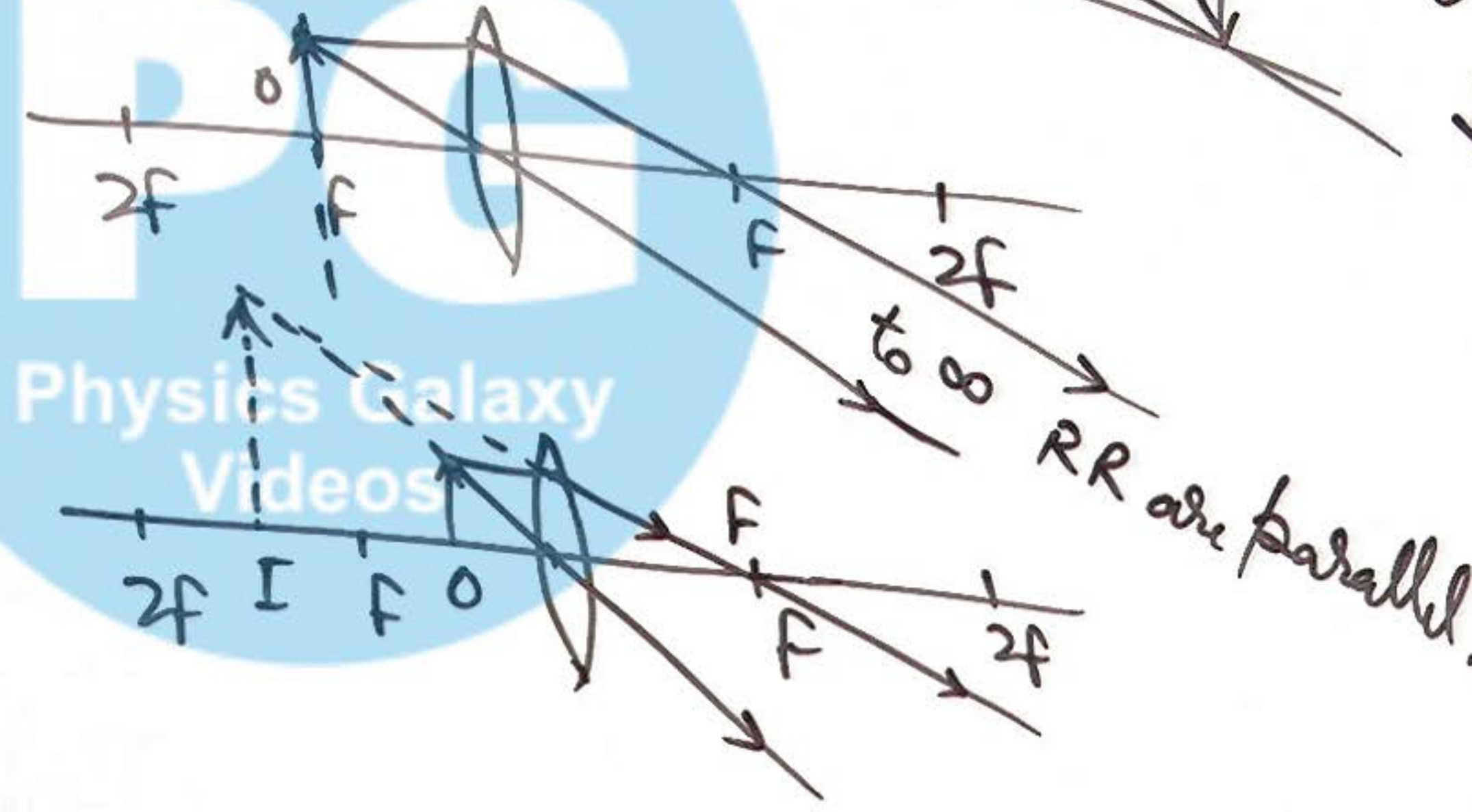
Object is beyond $2f$



Real.
I betwⁿ F & $2f$
Smaller in size



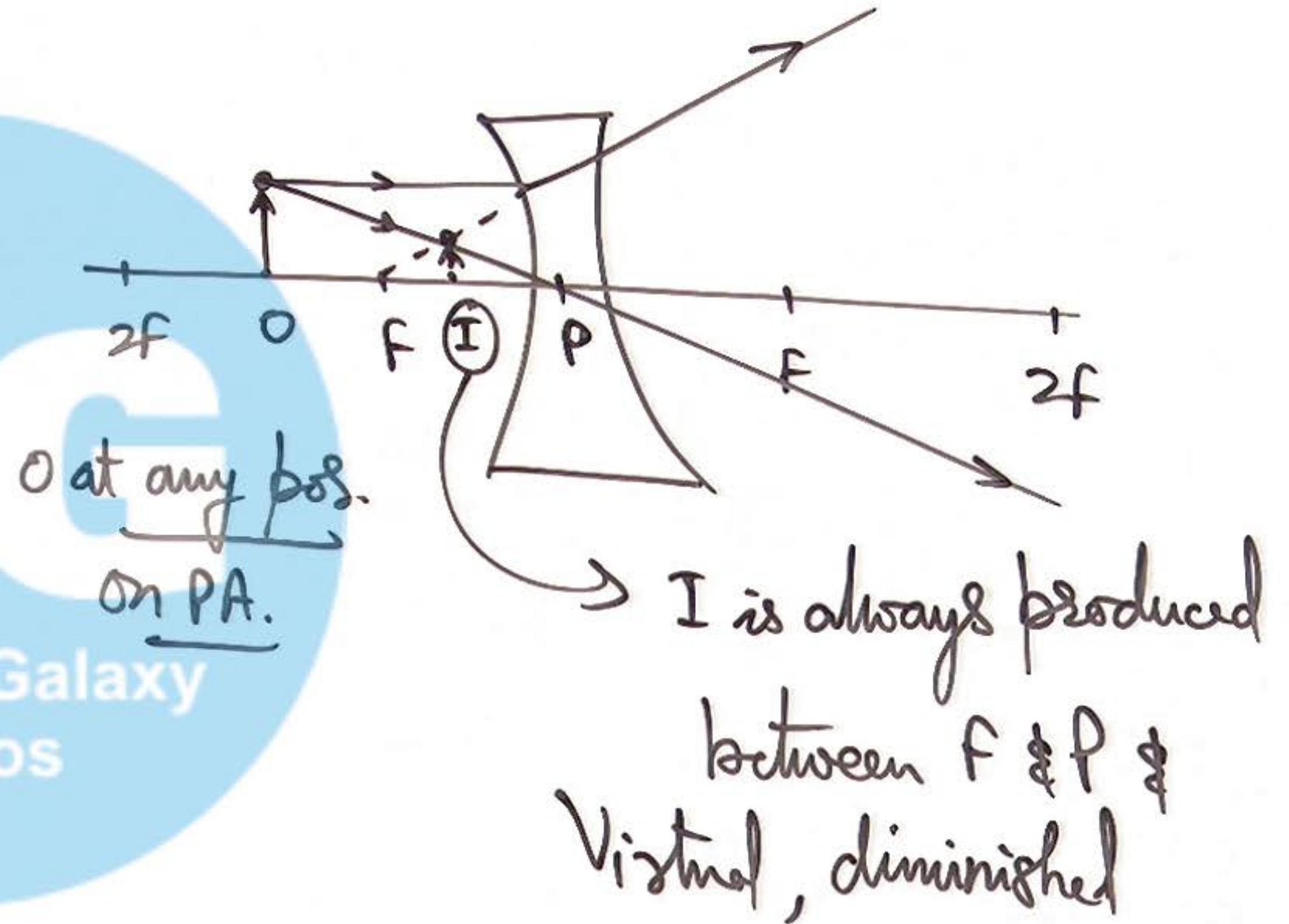
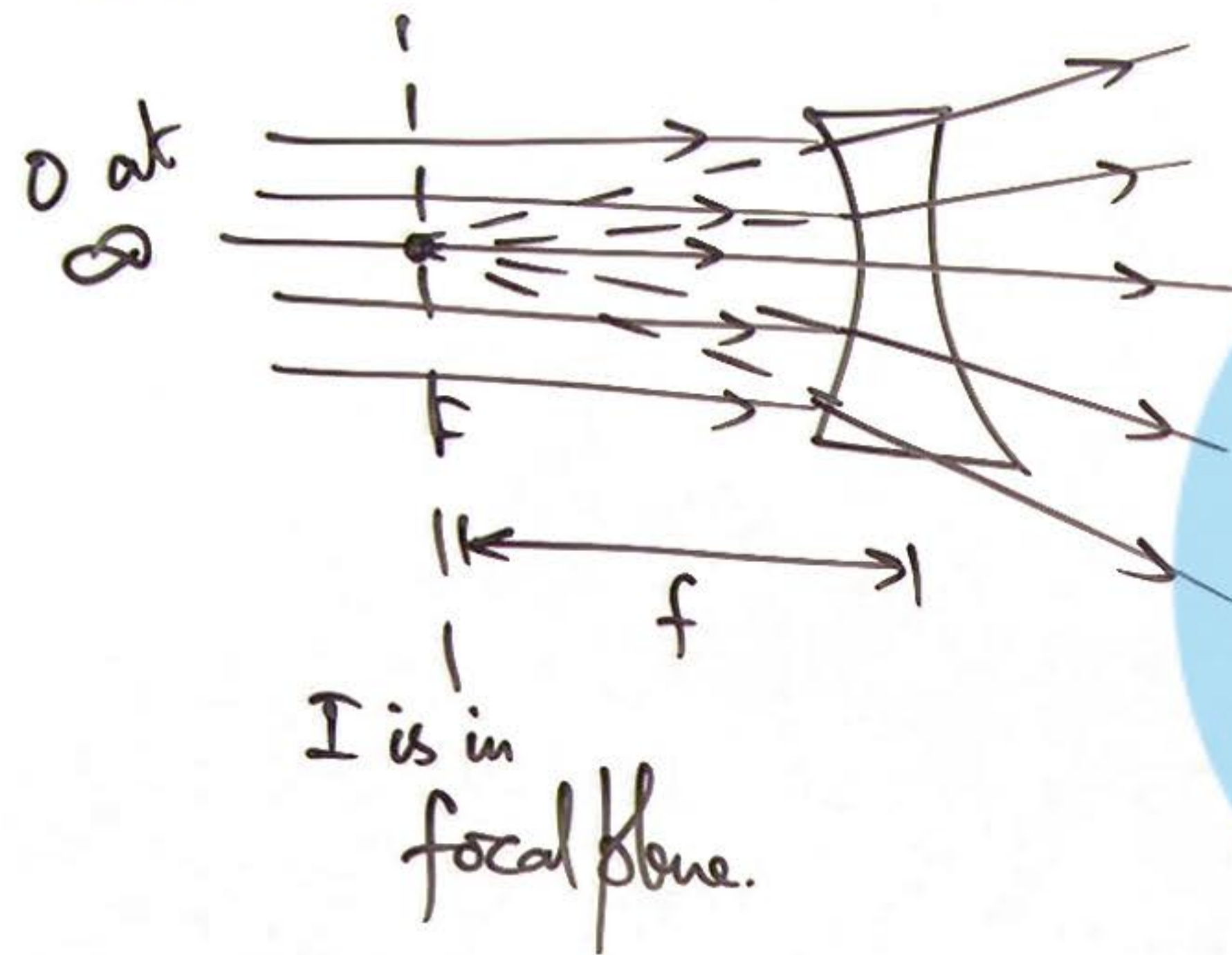
Enlarged
↓ Real.



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④ Relative position of Images produced by Thin Lenses:

Case-II:
Concave lenses



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5) Focal length of a Thin lens:

Lens maker's Formula.

Equiconvex/Equiconcave lens.

for BiConvex/Biconcave

$$\left| \frac{1}{f} \right| = (\mu - 1) \left[\frac{R_1 + R_2}{R_1 R_2} \right]$$

$$\Rightarrow \boxed{\left| f \right| = \frac{R_1 R_2}{(\mu - 1)(R_1 + R_2)} = \frac{R}{2(\mu - 1)}}$$

RI of lens w.r.to medium

$$\boxed{\frac{1}{f} = (\mu - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]}$$

for surr med RI = μ_s

$$\frac{1}{f} = \left(\frac{\mu_L}{\mu_s} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

when
 $R_1 = R_2 = R$

For plano Convex/plano Concave

$$\left| \frac{1}{f} \right| = (\mu - 1) \left(\frac{1}{R} \right)$$

$$\Rightarrow \boxed{\left| f \right| = \frac{R}{\mu - 1}}$$

for Concavo Convex/Convexo Concave

$$\boxed{\left| f \right| = \frac{R_1 R_2}{(\mu - 1) |R_1 - R_2|}}$$

⑥ Analysis of Image Formation by Thin Lenses:

Parameters for image produced by thin lenses:

(i) Location of Image $\rightarrow$ lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

(ii) Nature of Image $\rightarrow$ Converging RL $\rightarrow$ Real
Diverging RL $\rightarrow$ Virtual

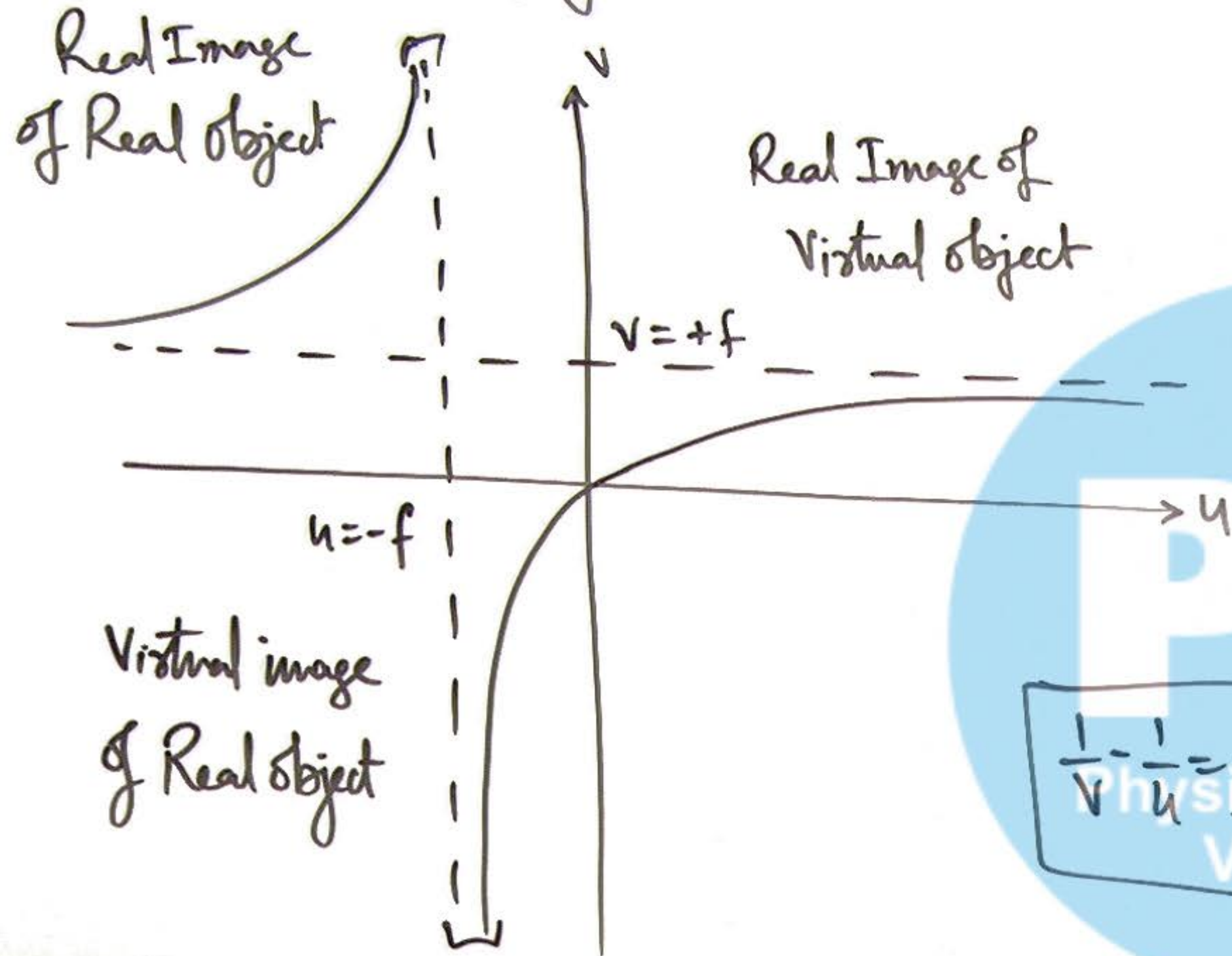
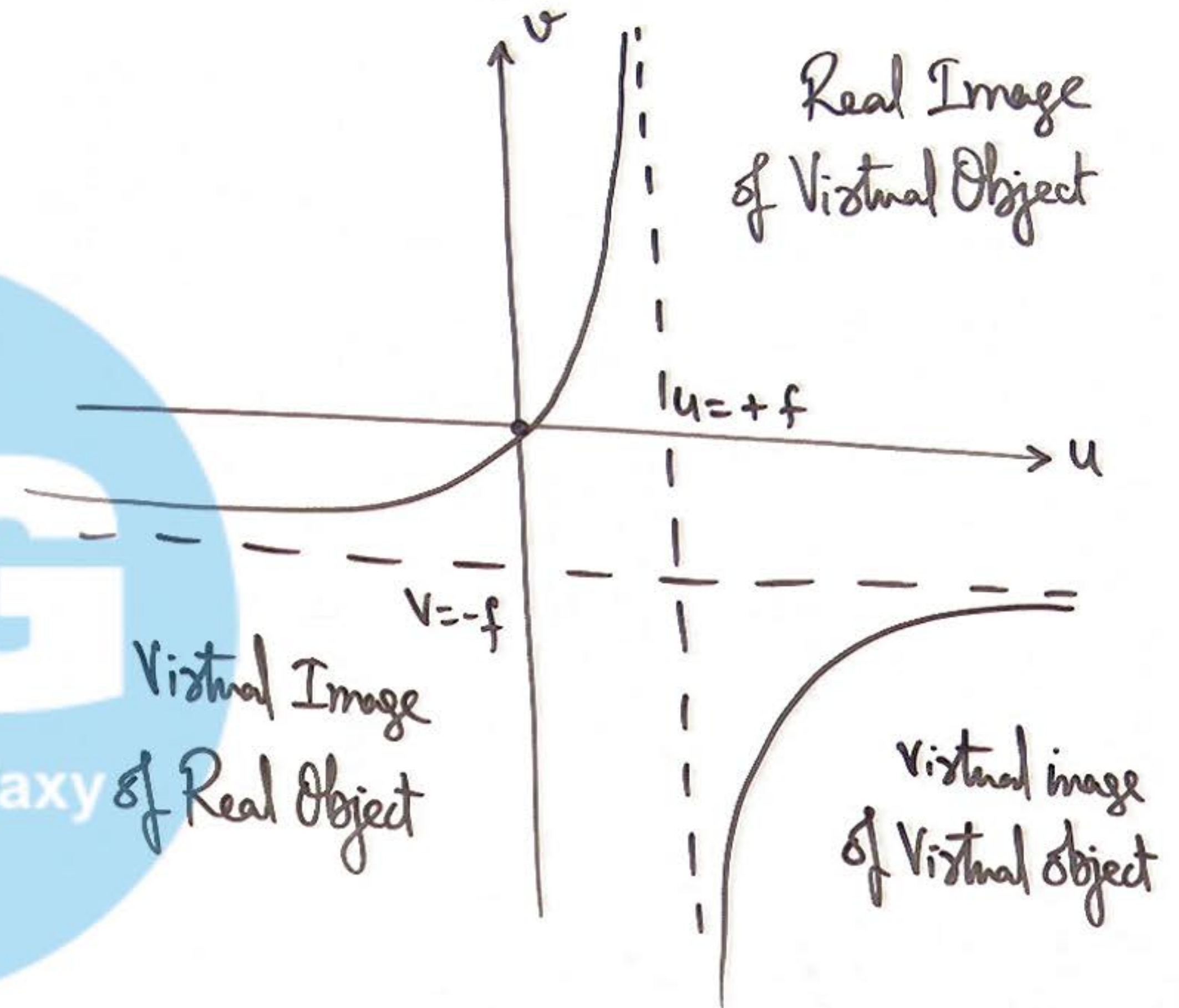
(iii) Size of Image $\rightarrow$ Magnification formula

$m = \left| \frac{v}{u} \right|$ (lateral magnification) $\rightarrow$ velocity magnification

$m_l = \frac{v^2}{u^2}$ (longitudinal mag)

✓ (iv) Orientation of Image

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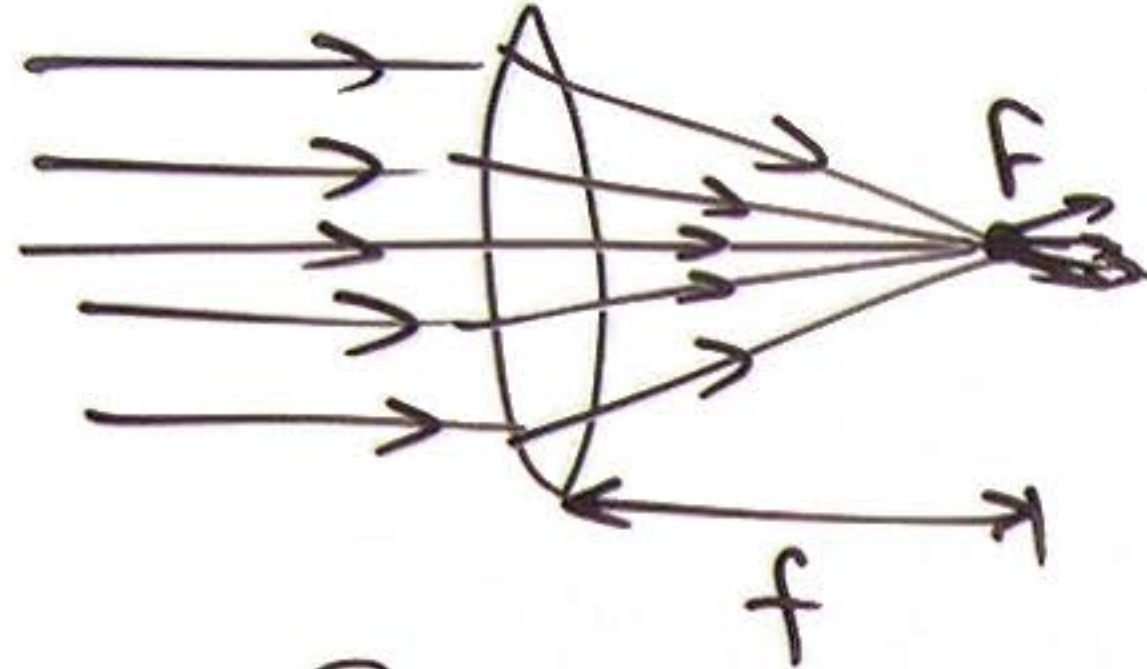
⑦ Variation Curves of Object & Image distances :Converging lenses ($f = +ve$)Diverging Lens ($f = -ve$)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

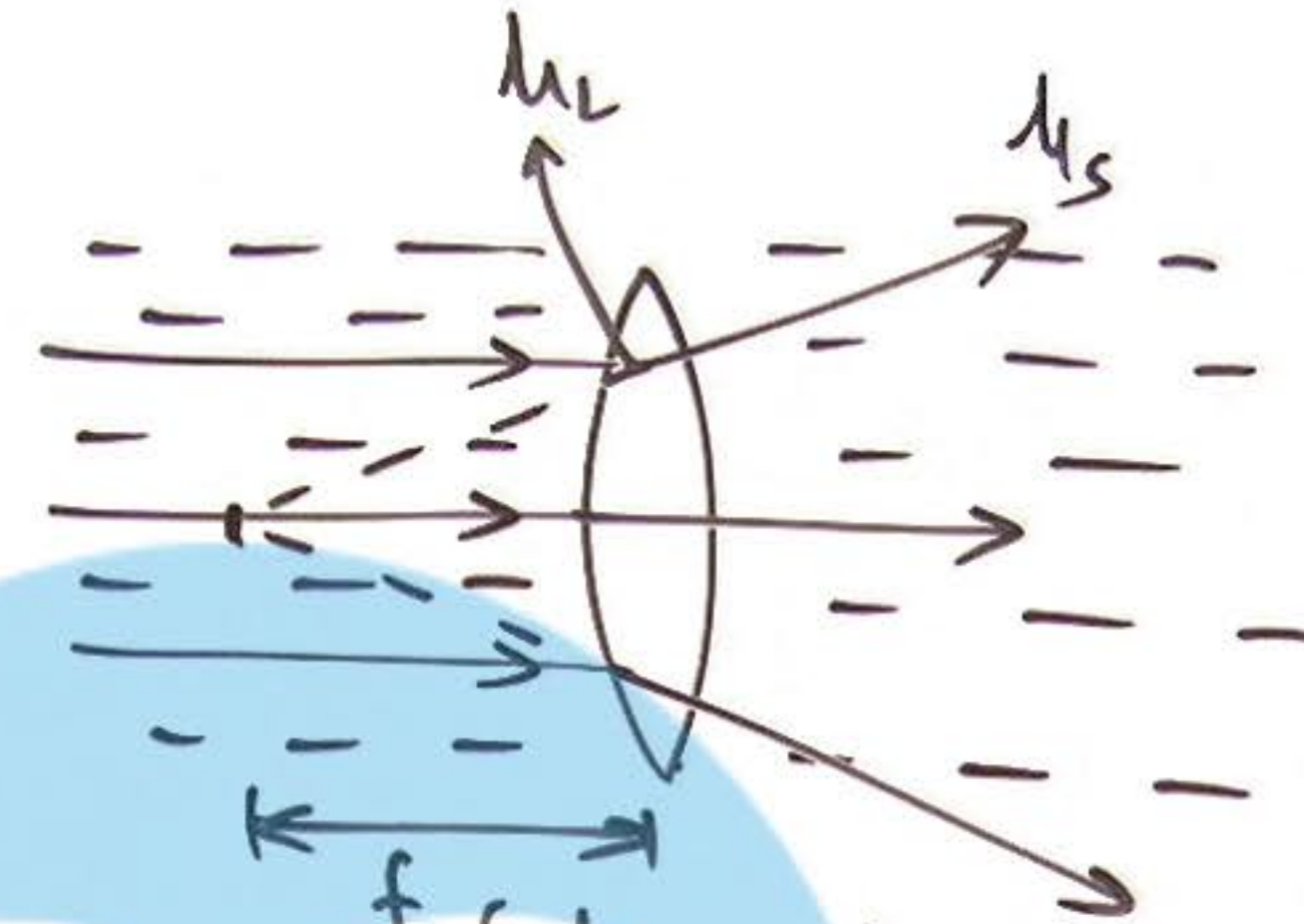
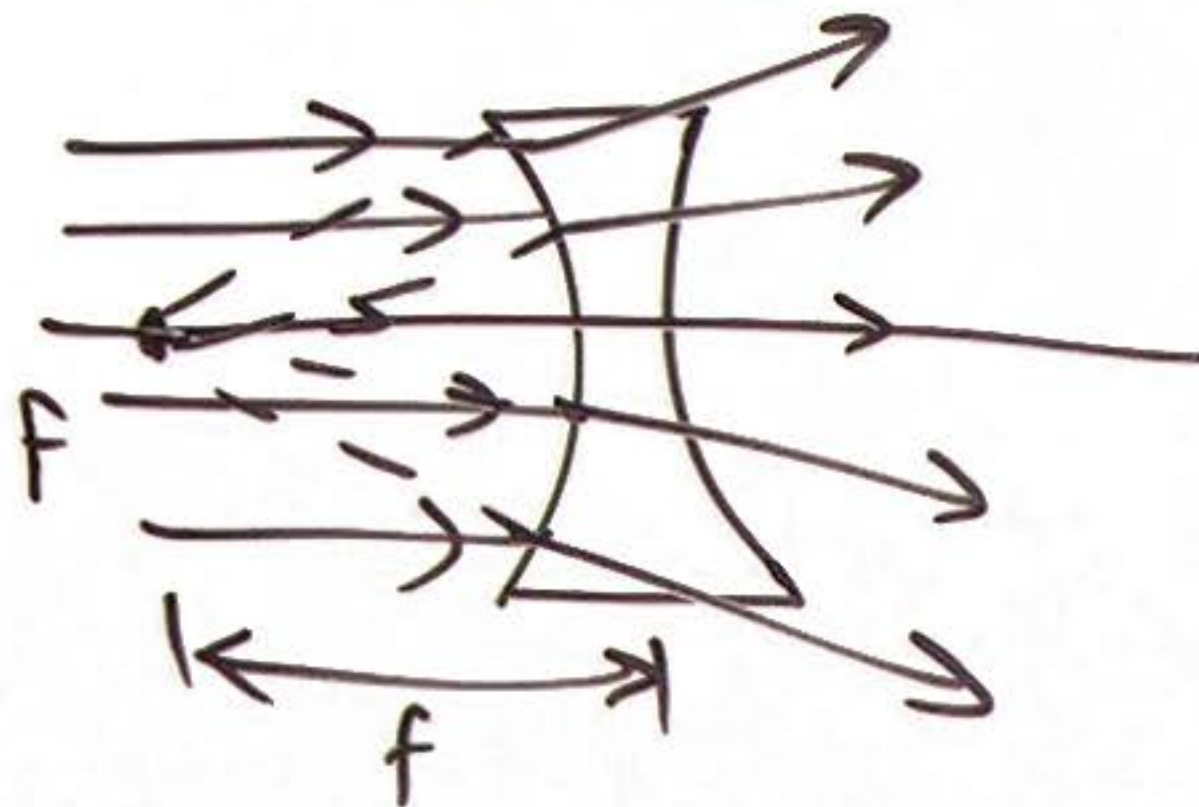
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⑧ Converging & Diverging Nature of Thin Lenses:

Convex



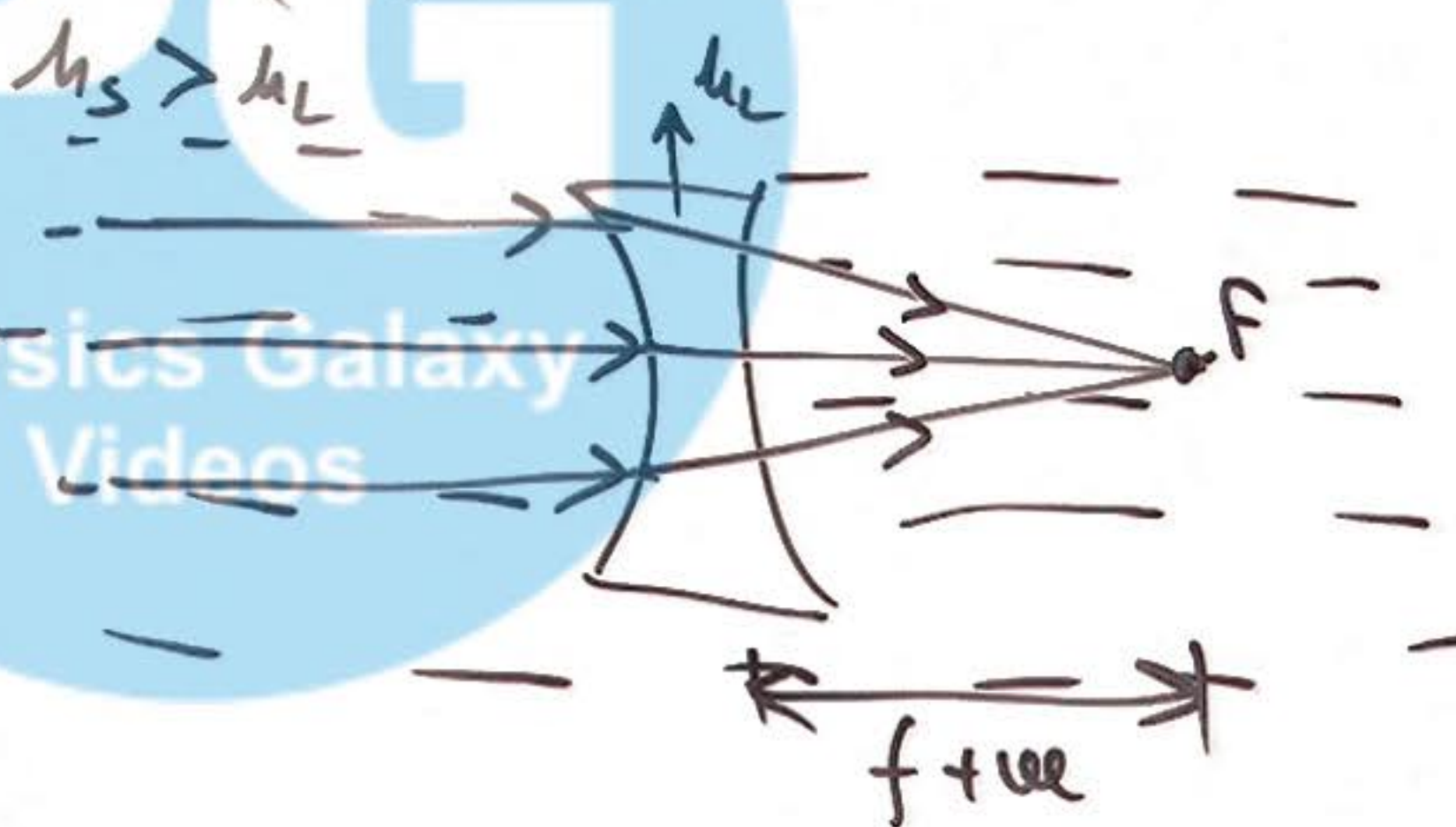
Concave



$(-ve)$ ($h_S > h_L$)

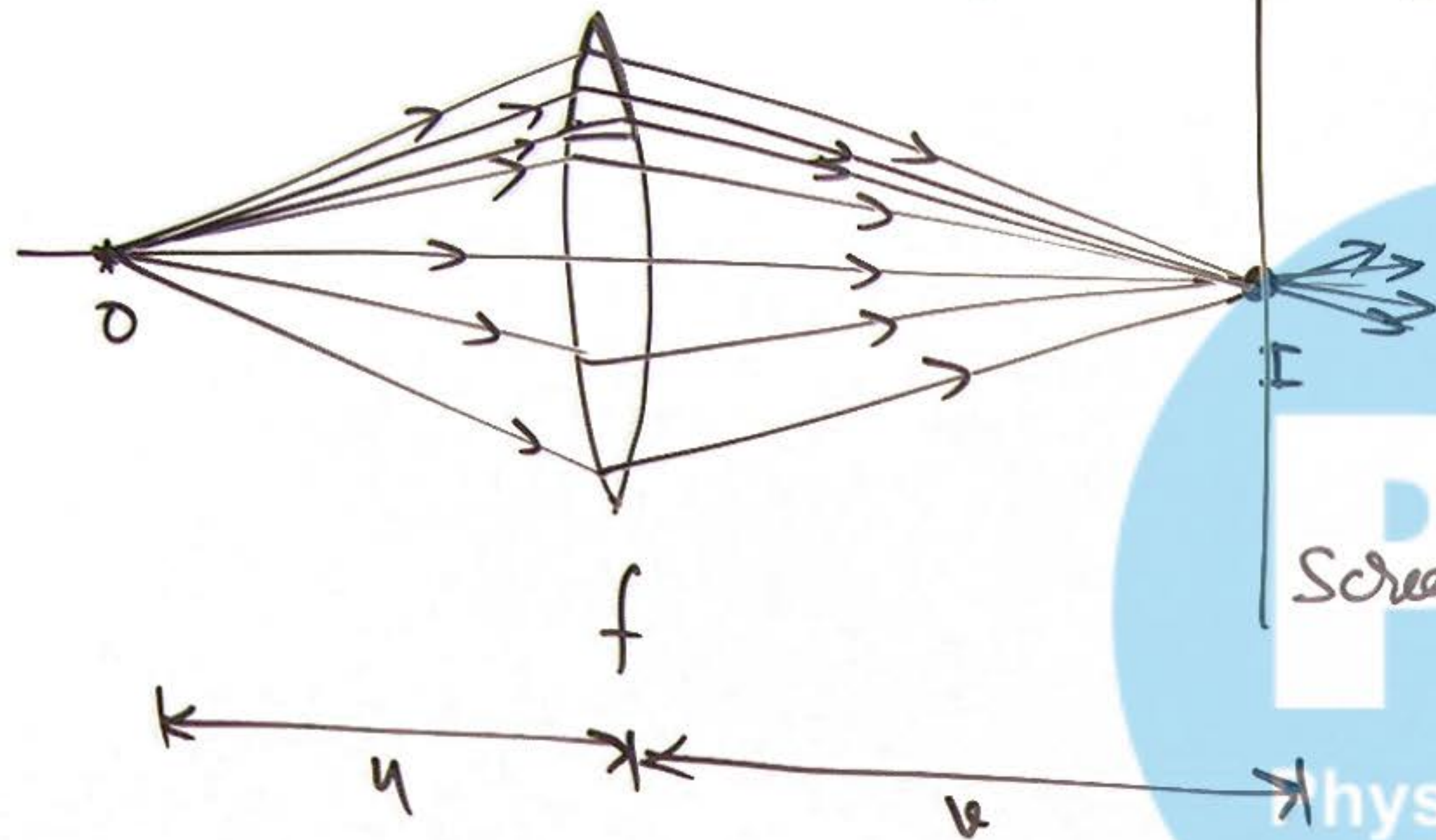
$h_S > h_L$

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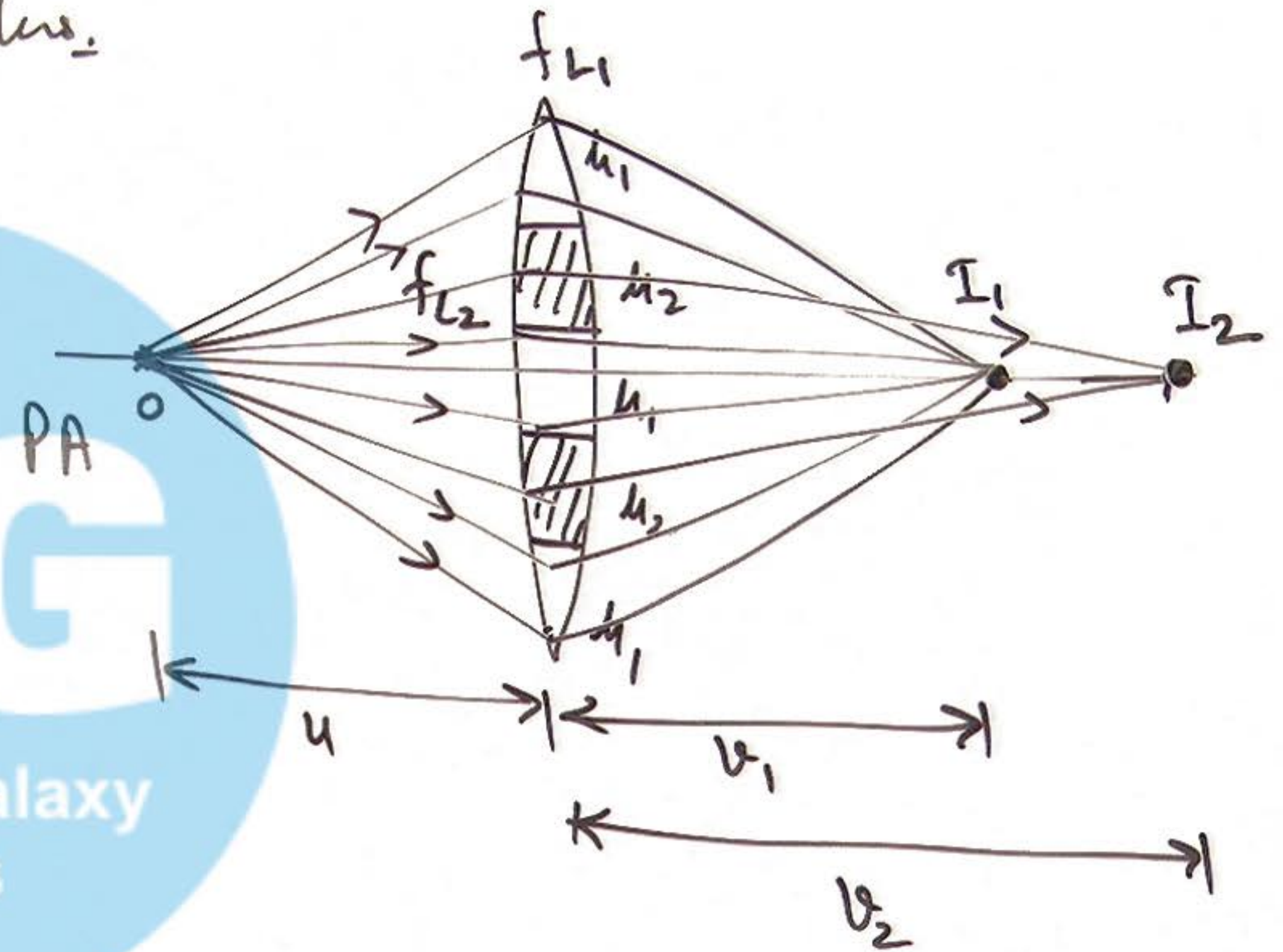


⑨ Effect of Cutting a Lens:

Every small part of lens
Behaves like complete lens.



$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$



⑩ Optical Power & Lens-Mirror Combination :

Optical Power

$$P = \frac{1}{f}$$

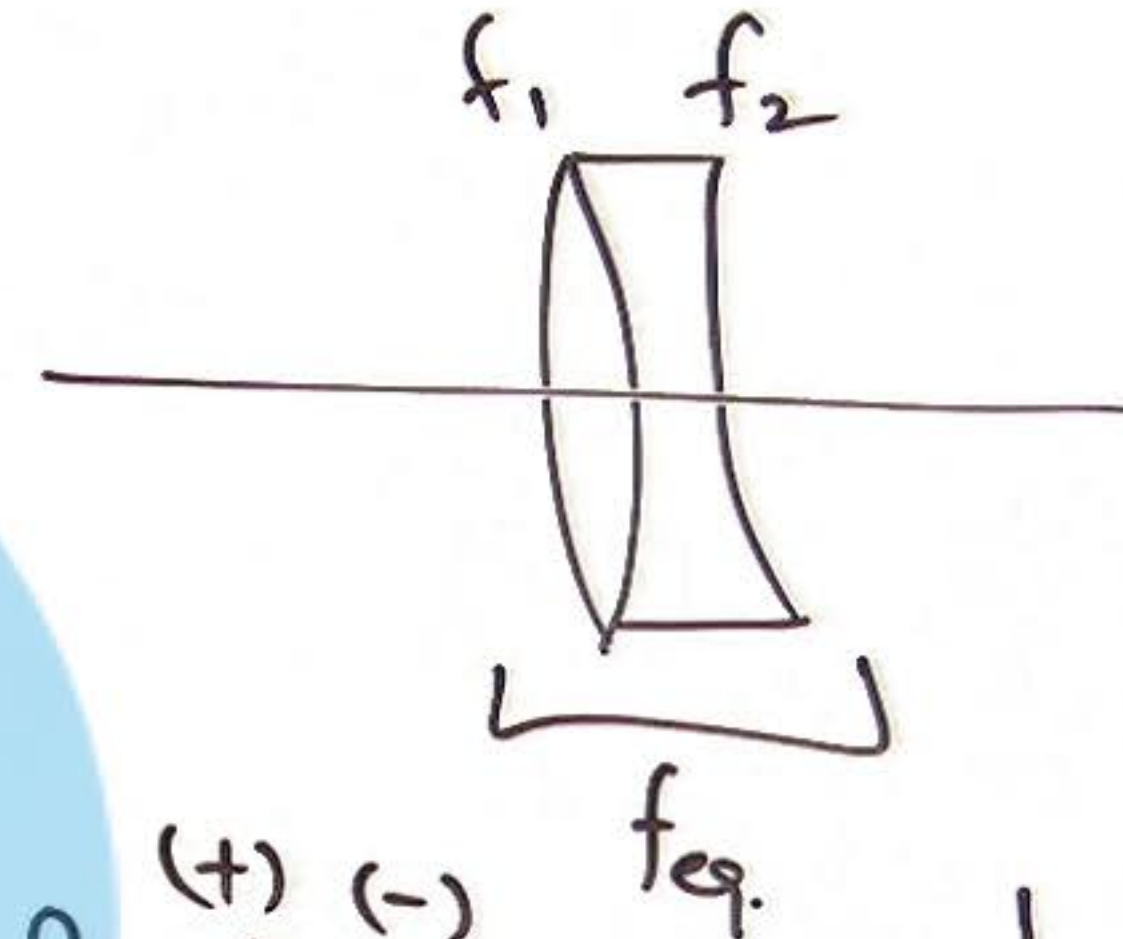
Powers are added up.

if f is in 'm'

then $P \rightarrow$ unit 'dioptre'
"D"

for Converging device $P \rightarrow +ve$
for Diverging devices $P \rightarrow -ve$

Lens Combination



Lens & Mirror Combination



$$P_{eq} = P_1 + P_2 \rightarrow \frac{1}{f_{eq}} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$P_{eq} = P_L + P_M + P_L \Rightarrow \frac{1}{f_{eq}} = \frac{1}{f_M} + \frac{2}{f_L}$$

⑪ Image formation at location of object:

IL: find h so that image of pin is produced on itself

use LMF

$$f_L = \frac{R_1 R_2}{(n-1)(R_1 - R_2)}$$

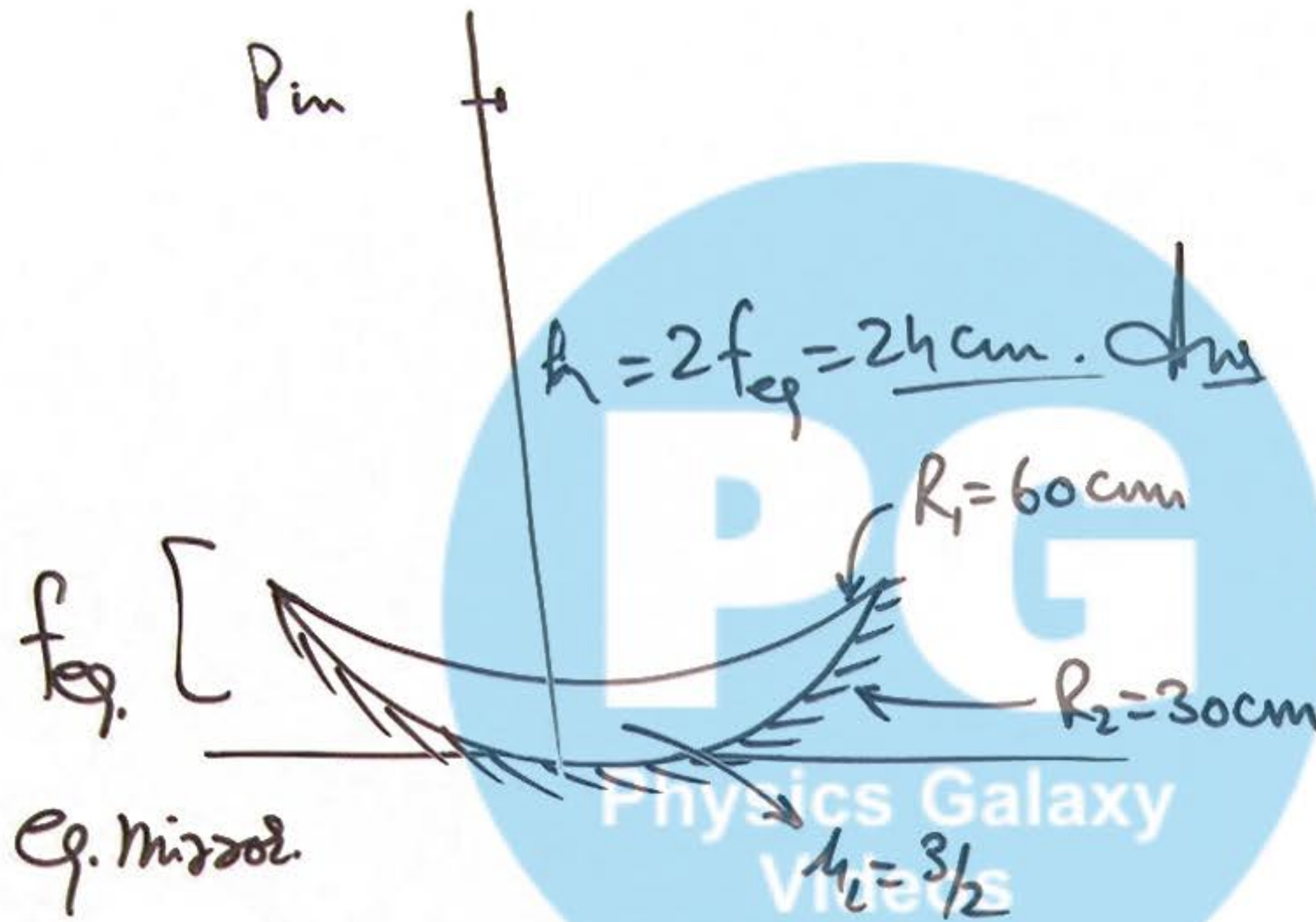
$$f_L = \frac{3/0 \times 60}{0.5 \times 30} = 120 \text{ cm}$$

$$f_M = \frac{R_2}{2} = 15 \text{ cm}$$

$$\frac{1}{f_{eq}} = \frac{1}{f_M} + \frac{2}{f_L}$$

$$\frac{1}{f_{eq}} = \frac{1}{15} + \frac{2}{120}$$

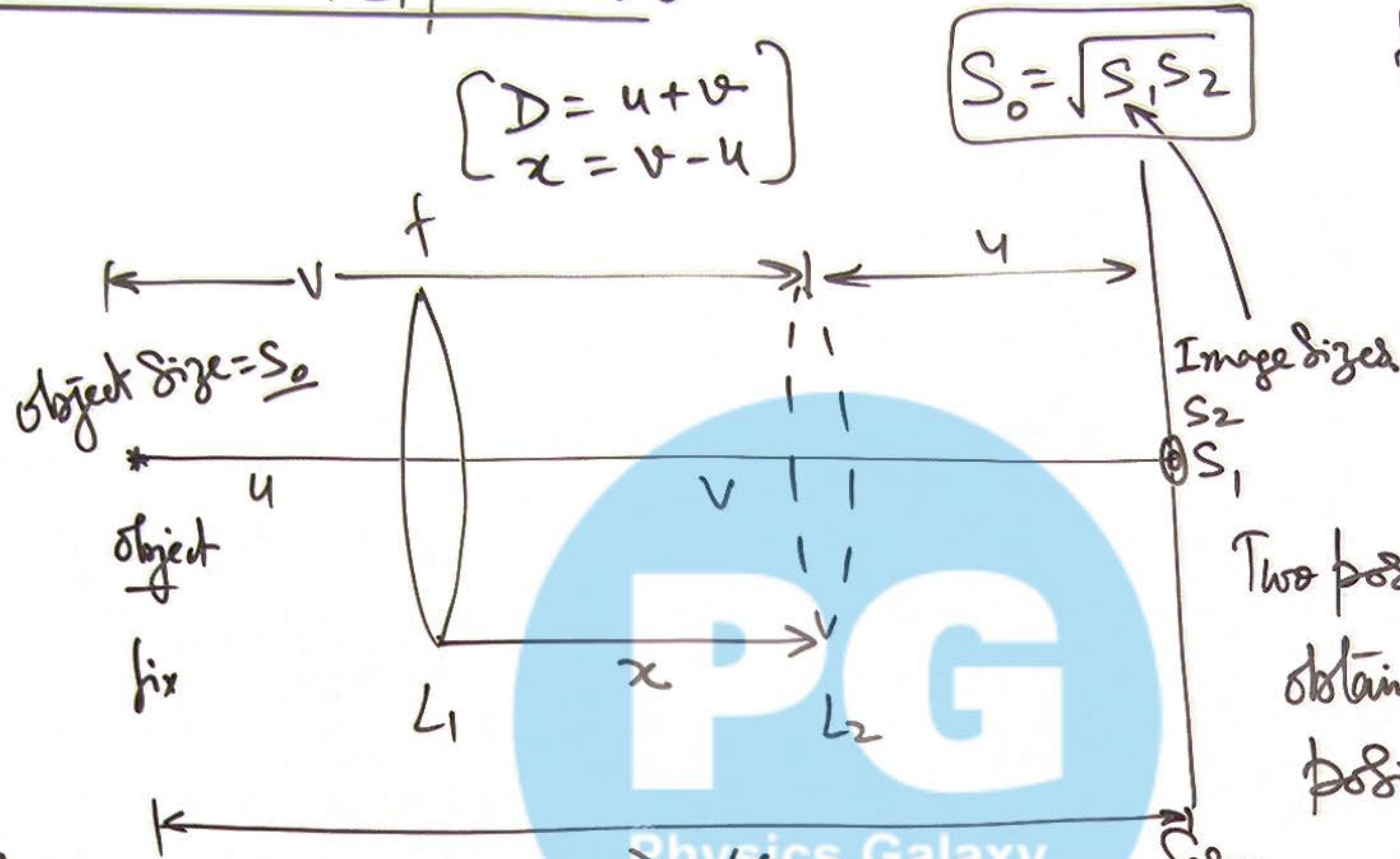
$$= \frac{5}{60} \Rightarrow f_{eq} = 12 \text{ cm.}$$



12 Displacement Method Experiment:

for obtaining
real image of
a real object
by a convex
lens separation
between O & I
must be

$$L > 4f$$



$$\begin{cases} D = u + v \\ x = v - u \end{cases}$$

$$S_o = \sqrt{S_1 S_2}$$

by lens formula.

$$\frac{1}{v} - \frac{1}{-u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

Two possible sharp images can be obtained on screen for two lens positions.

focal length of lens

$$f = \frac{D^2 - x^2}{4D}$$

$$\begin{cases} \text{for } L_1, m_1 = |v/u| \\ \text{for } L_2, m_2 = |u/v| \end{cases} \Rightarrow m_1 m_2 = 1$$

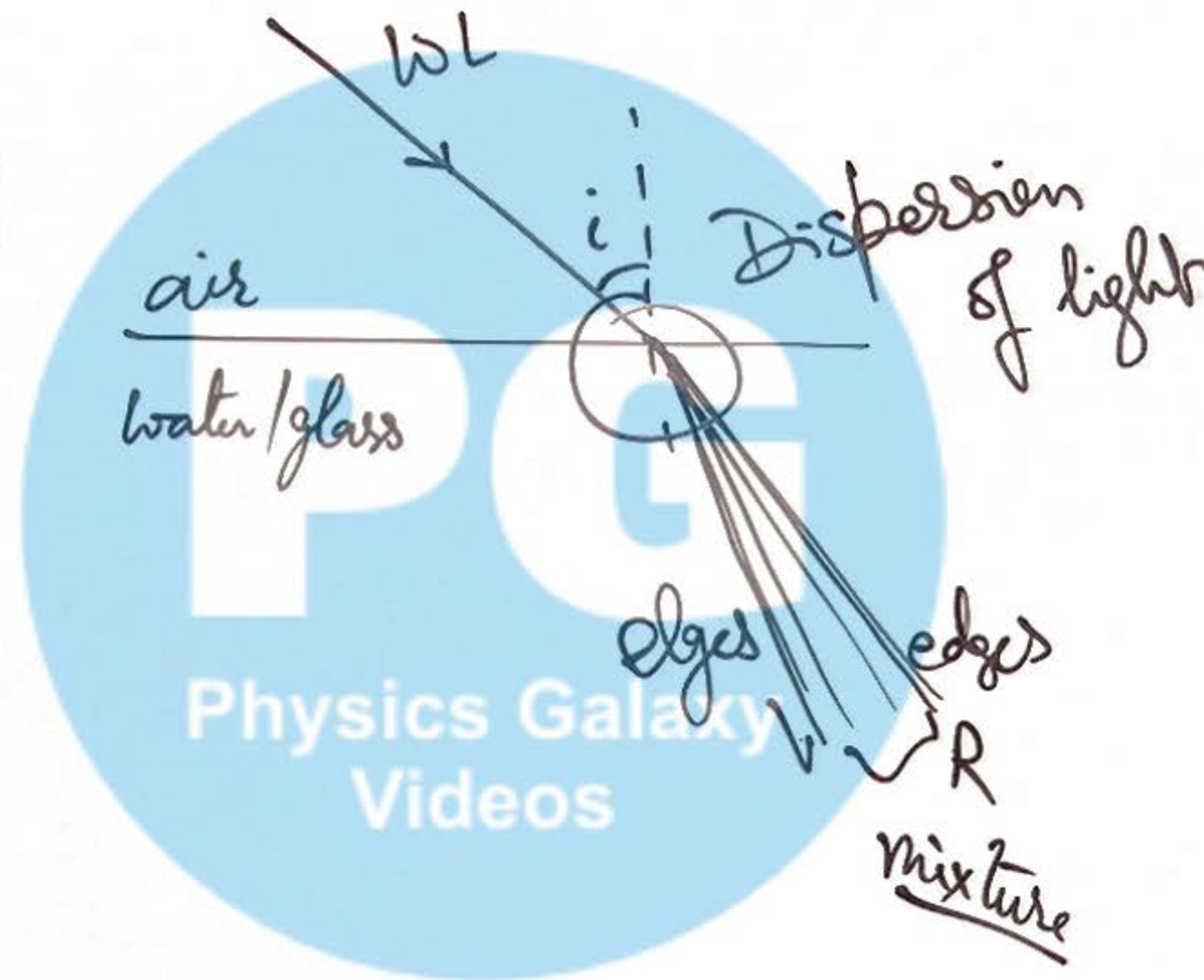
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13 Dispersion of Light:Cauchy's Relation

$$\mu = A + \frac{B}{\lambda^2}$$

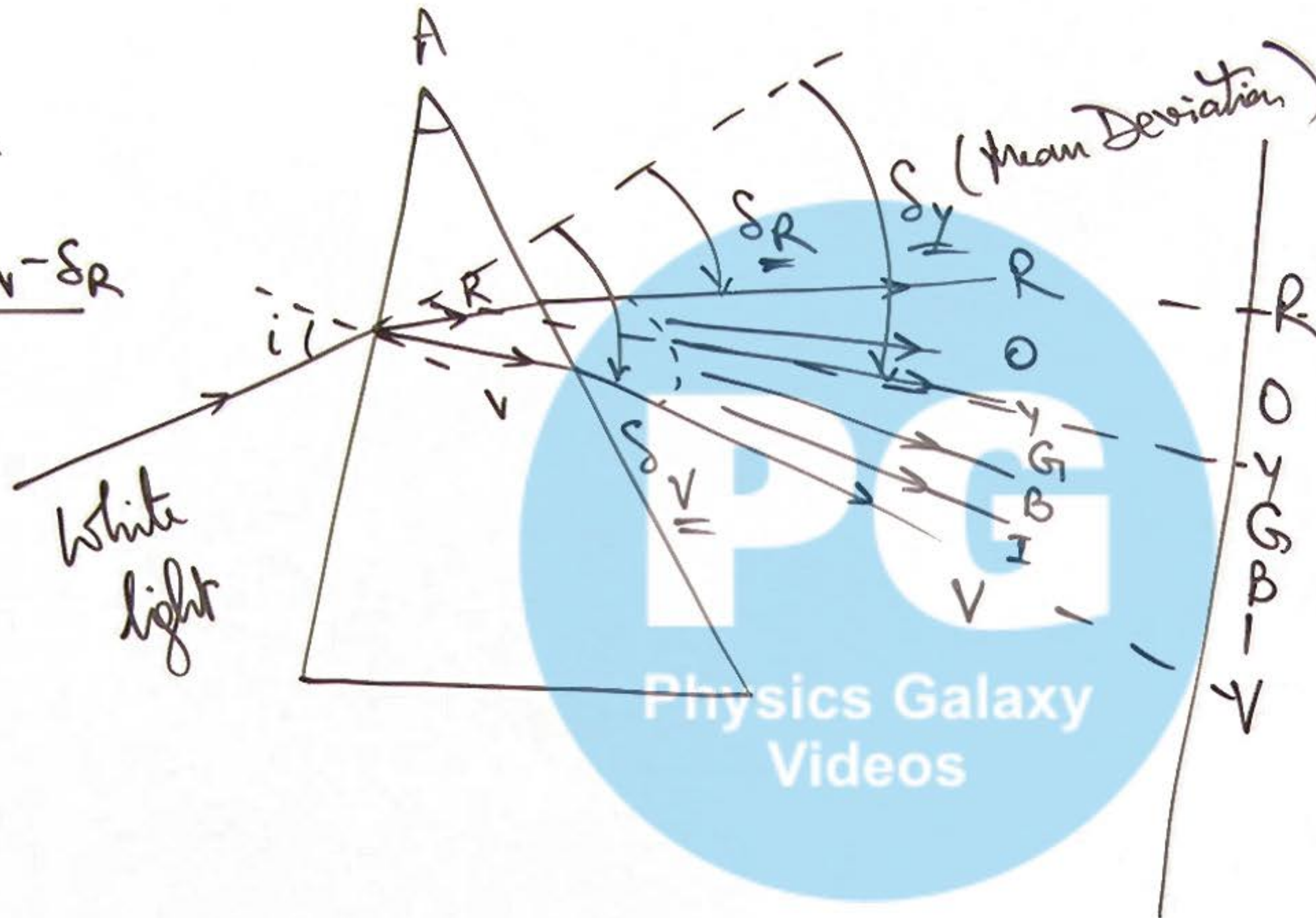
$$\text{As } \lambda_R > \lambda_V$$

$$\Rightarrow \underline{\underline{\mu_R < \mu_V}}$$



14) Dispersion Through a Prism:

Dispersion
Angle $D = \delta_v - \delta_R$



15) Dispersion due to Small angled Prism:

$$\delta_R = A(\mu_R - 1)$$

$$\delta_v = A(\mu_v - 1)$$

$$\delta_y = A(\mu_y - 1)$$

16) Dispersive Power of a material (Prism)

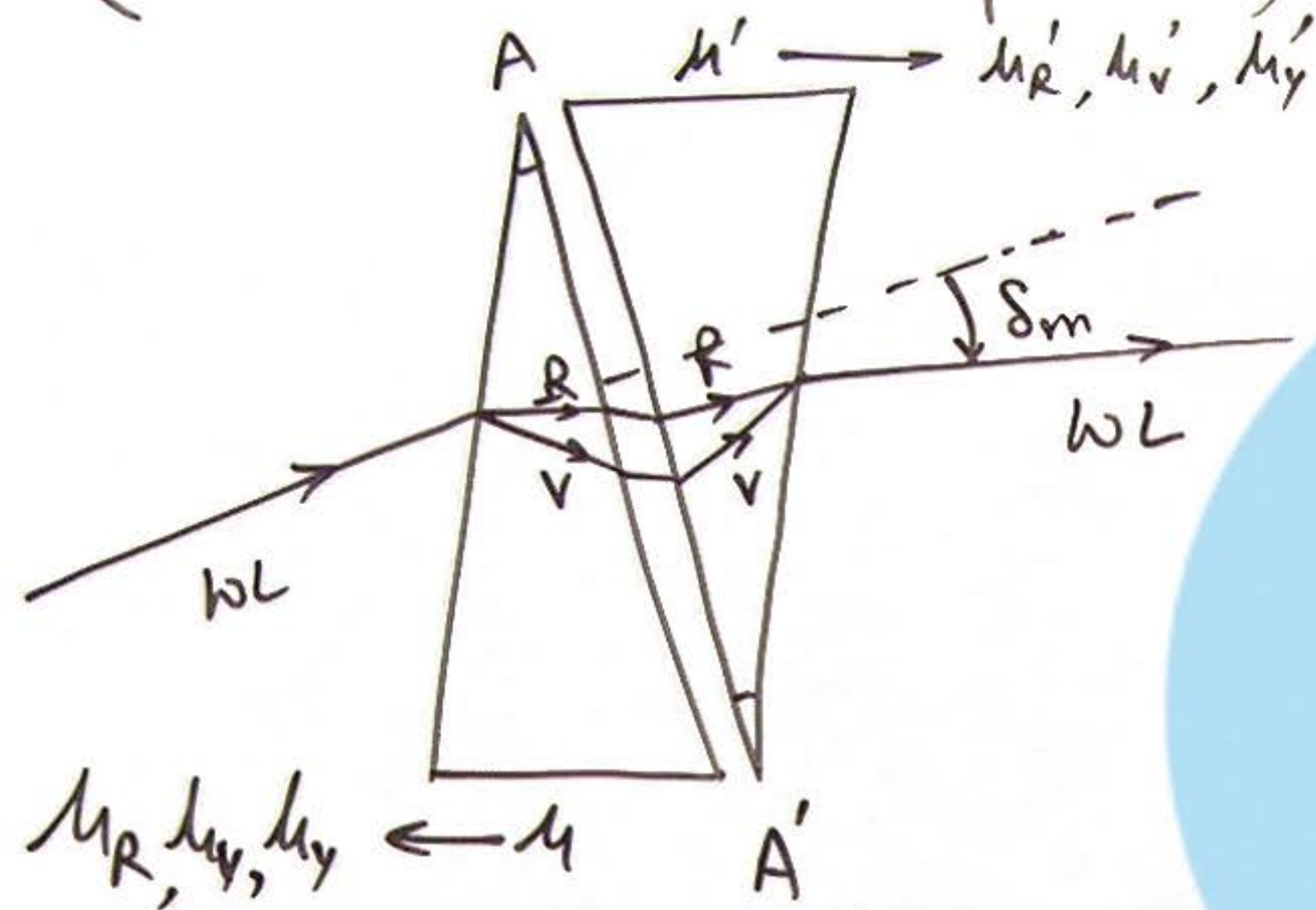
$$\omega = \frac{d\delta}{\delta} = \frac{\Delta\delta}{\delta} = \frac{\mu_v - \mu_R}{\mu_y - 1}$$

$\Delta\delta$ = Dispersion Angle
 δ = mean Deviation

17) Small Angled Prism Combinations:

(A) Achromatic Combination

(Deviation without Dispersion)

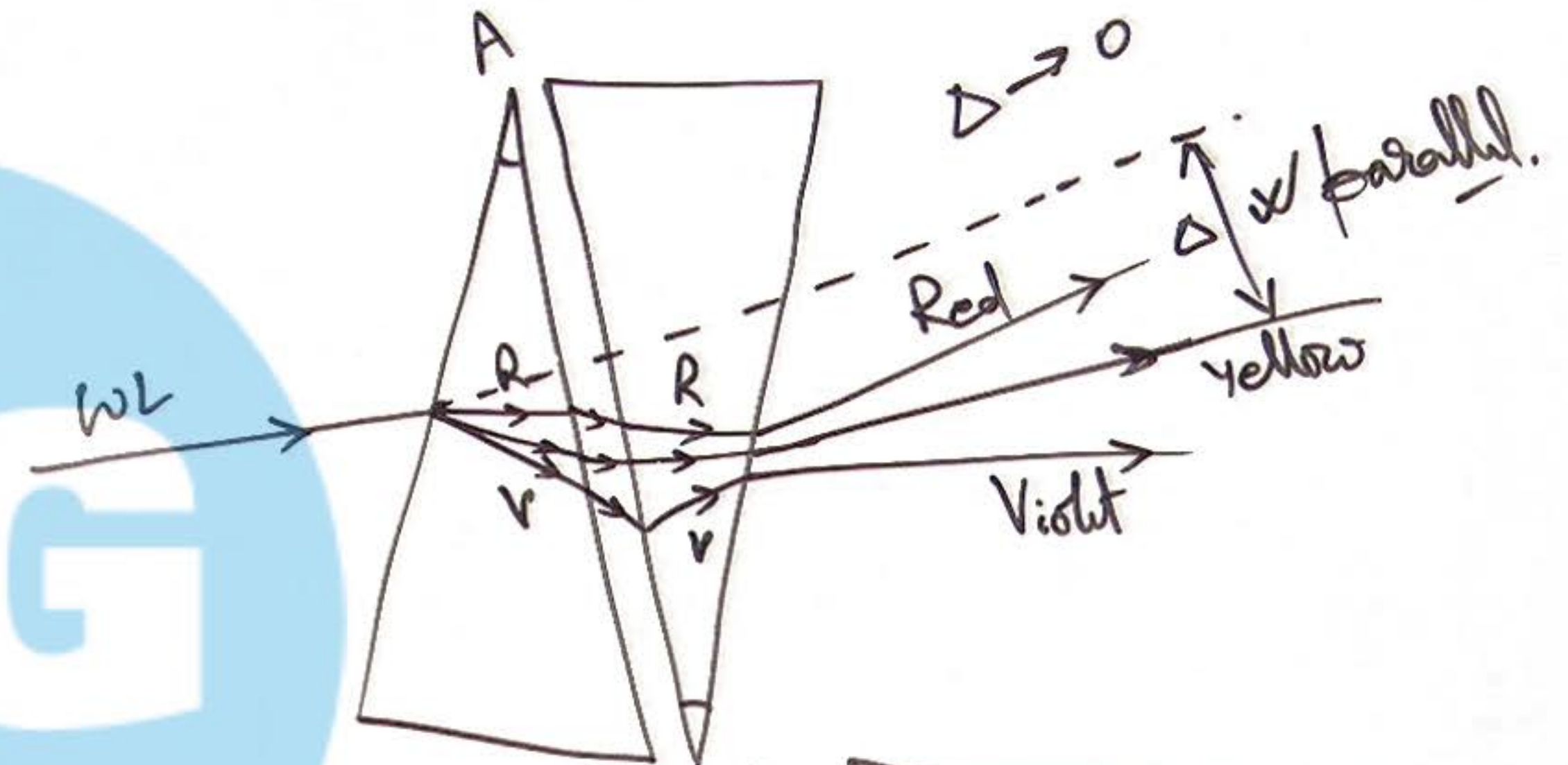


$$D = D'$$

$$\begin{aligned} \checkmark A(\mu_v - \mu_R) &= A'(\mu'_v - \mu'_R) \\ \checkmark \delta_m = \delta_1 - \delta_2 &= A(\mu_y - 1) - A'(\mu'_y - 1) \end{aligned}$$

(B) Direct Vision Combination

(Dispersion without Deviation)



$$\delta_m = \delta'_m \Rightarrow$$

$$A(\mu_y - 1) = A'(\mu'_y - 1)$$

Total Dispersion

$$D_T = D - D' = A(\mu_v - \mu_R) - A'(\mu'_v - \mu'_R)$$

18) Optical Aberrations in lenses & mirrors:

defects

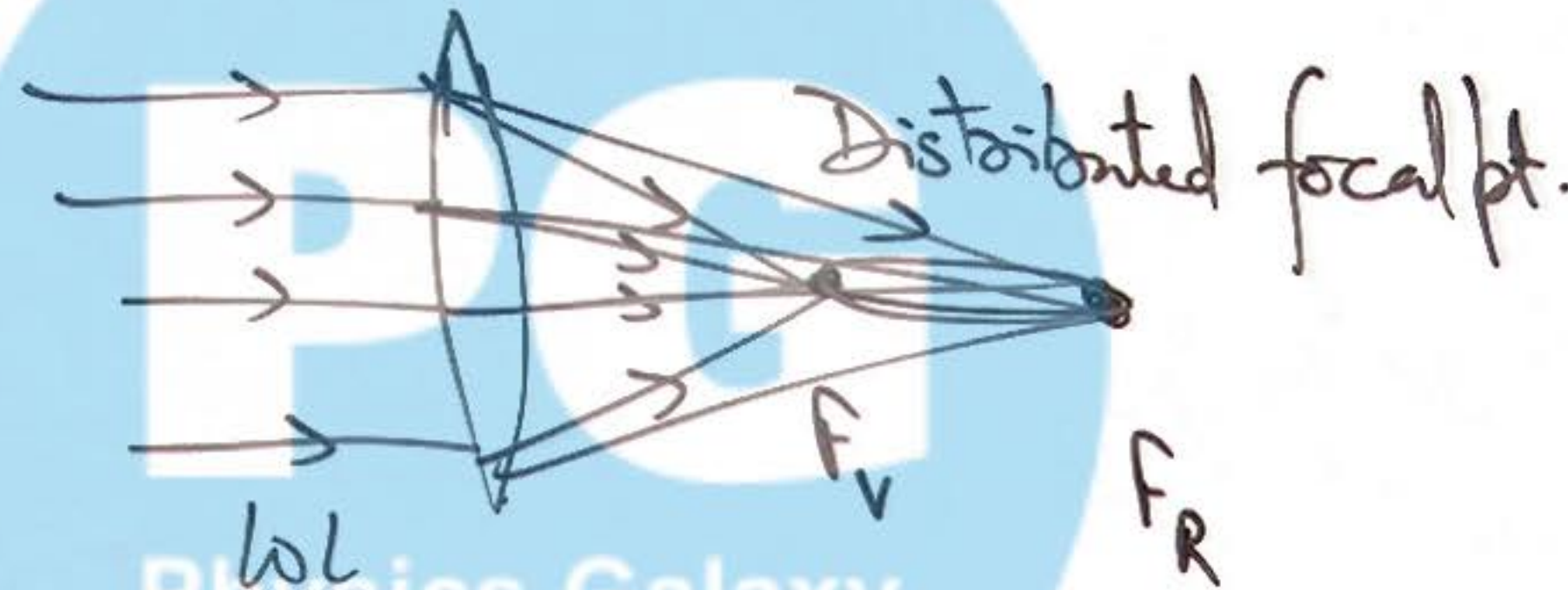
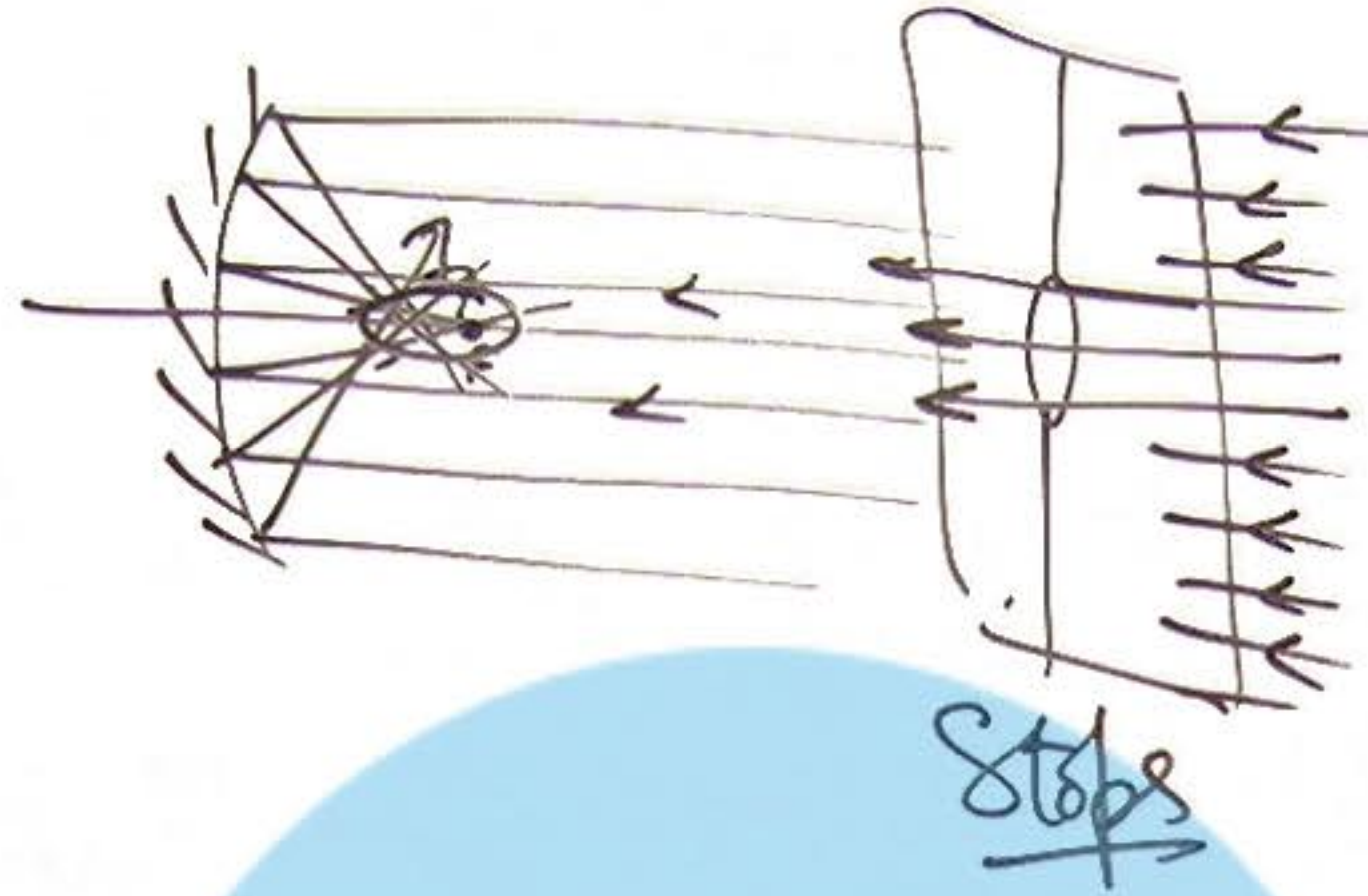
(i) Spherical aberrations

(ii) chromatic aberrations

λ

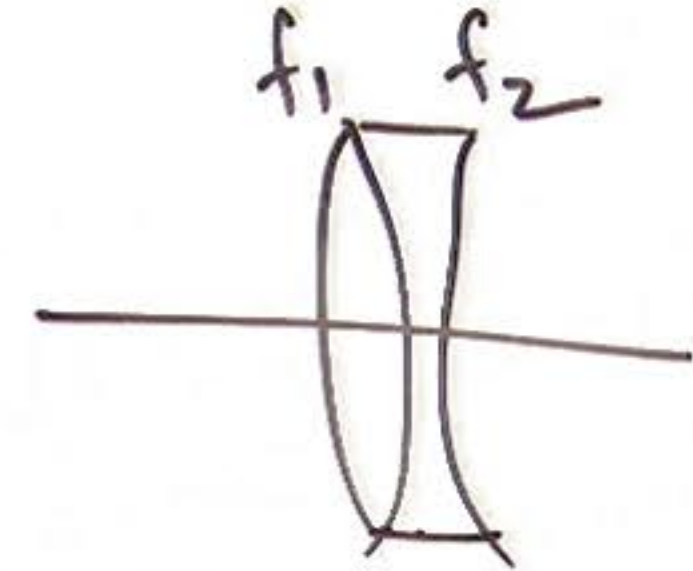
Mirrors are free from

chromatic aberration



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19) Achromatic Combination of lenses



$$\frac{\omega_1}{f_1} + \frac{\omega_2}{f_2} = 0$$